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UNIFORM SADDLEPOINT EXPANSIONS FOR MULTIVARIATE TAIL PROBABILITIES

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We extend the first-order saddlepoint approximation of Kolassa (2003) for the orthant probability $\mathbb{P}(\bar{X} \geq \bar{x})$ of the mean of n independent and identically distributed random vectors to an expansion of arbitrary order, and sharpen his error bound. The expansion is of Hauschildt–Robinson type: its leading term is a multivariate normal orthant probability times an exponential factor and its correction terms are derivatives of the normal orthant probability. The relative error after the terms of order $n^{-p/2}$ is $O(n^{-(p+1)/2})$, uniformly over compact sets of saddlepoints with nonnegative components, including those at which a component vanishes; saddlepoints with negative components are handled by an inclusion–exclusion identity as in Kolassa (2003), for which we give the closed form and conditions under which it preserves the error bound, and which we show cannot in general be dispensed with. The main contribution that makes the approximation usable at every order is that every term of every order is generated by two short recursions, one on the Taylor coefficients of the cumulant generating function at the saddlepoint and one for the derivatives of the normal orthant probability, in place of the enumeration of tensor invariants of the usual construction. The first recursion also gives the density expansion of any order, and the univariate expansions of Daniels and of Hauschildt and Robinson and the expansions for a vector of ratios are special cases. In bivariate and trivariate examples, the first-order approximations of Wang (1990), Kolassa (2003), and Kolassa and Li (2010) have relative errors of up to 7% in two dimensions and 27% in three; the terms of orders two to five reduce the error, at order five, to between 10^{-12} and 2×10^{-4}, except when $n = 1$.

1. Introduction. Let $X_1, \ldots, X_n$ be independent and identically distributed random d-vectors with cumulant generating function K, and let $\bar{X}$ be their mean. This paper considers the orthant probability $\mathbb{P}(\bar{X} \geq \bar{x})$, where $\bar{x}$ is a d-vector, from which the joint distribution function and the probabilities of rectangles follow, and its main result is a saddlepoint expansion of this probability to arbitrary order. In one dimension, the saddlepoint approximations of Lugannani and Rice (1980), Hauschildt (1969) and Robinson (1982) to the distribution function of a sample mean are among the most accurate approximations available in statistics: under regularity conditions their relative error is $O(n^{-1})$, uniformly on compact sets of saddlepoints (Jensen, 1995, Sections 2.2 and 3.3), and they typically give two to four correct digits even for very small n; see Reid (1988) and Barndorff-Nielsen and Cox (1989) for surveys of the univariate theory.

Theorem 1 extends the approximation of Kolassa (2003) to an expansion of any order. It is of Hauschildt–Robinson type: a multivariate normal orthant probability plus terms in the derivatives of the normal orthant probability, all multiplied by a common exponential factor.

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Unlike the Lugannani–Rice type approximations of Wang (1990) and Kolassa and Li (2010), it has no singular terms, its terms instead extending continuously to saddlepoints with vanishing components, and it does not depend on the order of the coordinates. The relative error after the terms of order $n^{-p/2}$ is $O(n^{-(p+1)/2})$, uniformly over compact sets of saddlepoints in the closed positive orthant. This bound holds up to and including saddlepoints with vanishing components, where the leading term is of larger order than in the interior, $n^{-(d-k)/2}$ rather than $n^{-d/2}$ times the exponential factor when k components vanish, so that a bound at the scale of the exponential factor like that of Kolassa cannot be relative on both sides of the boundary. Saddlepoints with negative components are handled, as in Kolassa (2003), by inclusion–exclusion, for which we give the closed form and conditions under which the relative error bound carries over to the reduced approximation (Theorem 2); we also show that this reduction is needed for a generally valid procedure, since the expansion applied directly at a saddlepoint with a negative component in general has the wrong exponent, its ratio to the true probability tending to zero or to infinity exponentially in n (Proposition 1), the exception being a profiled cumulant generating function that is quadratic in the negative direction.

In several dimensions the literature is sparser, and even the first-order approximations are comparatively recent: Wang (1990) for two dimensions, Kolassa (2003) and Kolassa and Li (2010) for general dimension; see Butler (2007, Section 3.5). For the multinomial, Dirichlet, multivariate hypergeometric and Pólya distributions, Butler and Sutton (1998) reduce the probabilities of rectangles to univariate saddlepoint approximations through conditional characterizations. The approximations of Wang (1990) and Kolassa and Li (2010) are of Lugannani–Rice type, built on signed roots of likelihood ratio statistics in the exponential family generated by tilting (Section 2). They depend on the order in which the coordinates are taken, and they have removable singularities at the points where a component of the saddlepoint or of the signed root vanishes, which need separate treatment and introduce numerical issues in the vicinity of the singularity; Kolassa and Li (2010) also note that the proof in Wang (1990) of the error rate near the pole is incomplete, and Kolassa (2003, Section 4) reports that a different argument gives the error rate in two dimensions but not beyond. The approximations of Wang (1990) and Kolassa and Li (2010) also evaluate the cumulant generating function away from the saddlepoint $\hat{s}$, at $(0, \hat{s}_2)$ and at $(\hat{s}_1, 0)$ respectively, and these points need not lie in its domain: for $X = Z + A$ and $Y = Z + B$, with Z standard Laplace and A, B standard normal, all independent, the saddlepoint of the point $(1.2, -1.2)$ is $\hat{s} = (1.2, -1.2)$, the domain is $|s_1 + s_2| < 1$, and neither approximation is defined in either order of the coordinates. The point is not in a remote tail, and the expansion of this paper is defined there (Section 5.2). The approximation of Kolassa (2003) is of Hauschildt–Robinson type: the exponent of the inversion integral is expanded and integrated term by term, which leads to derivatives of the normal orthant probability, and these are generated recursively up to the third order; the approximation uses the cumulant generating function only at the saddlepoint and at the saddlepoints of the marginals; its error bound is absolute, the exponential factor times $O(n^{-1})$, and Kolassa (2003, p. 279) points out that it is uniformly exponentially small but not, strictly speaking, both relative and uniform, a bound of that kind being harder to obtain for tail probabilities than for densities even in one dimension; Theorem 1 below provides one. The tail expansions of Withers and Nadarajah (2010a, Section 5) are of large-deviation type: the pole of the inversion integral is expanded in inverse powers of $n^{1/2}$ times the saddlepoint, so that their terms are singular when a component of the saddlepoint vanishes.

Importantly, every term of the expansion can be computed with ease: it is obtained by expanding $\exp\{nG(t)\} = \sum_{r=0}^{\infty} n^r G(t)^r / r!$, where G collects the terms of degree three and higher of the Taylor expansion of the cumulant generating function at the saddlepoint, and integrating term by term. The coefficients of the powers $G^r/r!$ are generated by one short

recursion on the Taylor coefficients, in any dimension, and the term-by-term integrals are derivatives of the multivariate normal orthant probability, generated by a second recursion (Lemma 2). Once the coefficients are organized by the excess of the degree over twice the number of factors r, which determines the order in n, every term of every order is produced by the same short program.

The density of $\bar{X}$ has the same structure, with moments of the normal distribution in place of the derivatives of the orthant probability, and Theorem 3 gives its expansion of any order from the same coefficients. The density expansion is the tilted Edgeworth expansion at the mean of the tilted distribution, which Kolassa (2006, Section 6.4) writes formally to all orders and Withers and Nadarajah (2010a, Section 5) give with coefficients expressed through Bell polynomials; in several dimensions, however, its terms have been evaluated only to the second order, in the tensor notation of Barndorff-Nielsen and Cox (1979) and McCullagh (1987), in which each additional order requires the enumeration of all invariants of the next degree. The recursion replaces this enumeration: the second-order term reproduces the invariants of Barndorff-Nielsen and Cox (1979) and McCullagh (1987), and the terms of higher order, which to our knowledge have not been computed before in several dimensions, are obtained by the same algorithm.

The univariate expansions and the expansions for ratios are special cases. For $d = 1$, Theorem 3 is Daniels' expansion and Theorem 1 is the expansion of Hauschildt (1969) and Robinson (1982), written with all of its terms explicit in Section 4, because the reduction of Section 3.3 needs the univariate expansions for its marginal terms; the expansion of Lugannani and Rice (1980) does not follow from Theorem 1, and we relate it to the Hauschildt–Robinson expansion in a remark, with the details in the Supplementary Material (Broda and Kan, 2026). The expansions for ratios, which the rest of the paper does not use, are worth recording and are relegated to Appendix A: the joint distribution function of a vector of ratios $\bar{X}_j/\bar{Y}_j$ is an orthant probability at the origin, to which Theorem 1 applies directly, and the joint density is a saddlepoint integral with an amplitude, for which Theorem 4 gives the expansion of any order; for $d = 1$ it is the expansion of Rice (1968), and for $d \geq 2$ it extends the first-order approximation of Butler and Paolella (1998) and Butler (2007, Section 12.3).

A number of applications illustrate the gains from the higher-order terms. We revisit the bivariate examples of Wang (1990) and Kolassa and Li (2010) and the trivariate example of Kolassa and Li (2010), with exact values in place of their simulations: the first-order approximations have relative errors of up to 7% in two dimensions and of 13–27% in three. The terms of orders two to five reduce these errors to at most 1.4×10^{-2} at order two and 2×10^{-4} at order five, and to about 10^{-7} or less at order five when $n \geq 10$. We then return to the Laplace example above, on which the two approximations are undefined. Next, we consider a ratio of regression coefficients, from a generalized confidence interval in a Fieller–Creasy problem, whose denominator takes both signs. For pairs of quadratic forms in normal vectors we compare the bivariate expansions with exact values obtained for a class of pairs with a common component. The sample autocorrelations at two lags, a pair of ratios of quadratic forms outside the setting of the theorems, are treated in the Supplementary Material.

Section 2 sets up the notation and the coefficient recursion, Section 3 gives the expansions of the multivariate distribution function and density, Section 4 the univariate expansions for a mean as corollaries, Section 5 the applications, and Section 6 concludes. Appendix A gives the expansions for vectors of ratios and for a single ratio; the proofs and notes on the computation are in Appendices B and C.

2. Notation and the coefficient recursion. Let $X_1, \ldots, X_n$ be independent and identically distributed d-vectors, and let $K(s) = \log E\{\exp(s^\top X)\}$ be their cumulant generating function, finite on an open set $\mathcal{T}$ containing the origin, with gradient $K'(s) = \partial K(s)/\partial s$

and Hessian $K''(s) = \partial^2 K(s)/\partial s\,\partial s^\top$, and let $\bar X = \sum_{i=1}^n X_i/n$. Let $M_s(u) = \exp\{K(s + \mathrm{i}u) - K(s)\}$ be the characteristic function of the distribution tilted by $s \in \mathcal{T}$. The tilted distributions P_s, $\mathrm{d}P_s(x) = \exp\{s^\top x - K(s)\}\,\mathrm{d}P(x)$ with P the distribution of X, form the natural exponential family with canonical parameter s and cumulant function K; when $\bar x$ is the observed mean, the saddlepoint $\hat s$ defined below is the maximum likelihood estimate of s in this family, the saddlepoint equation being the score equation (Kolassa, 2003, p. 282), and the likelihood ratio and Wald statistics for $s = 0$ appear in Sections 3.3 and 4. We assume throughout that K'' is positive definite on $\mathcal{T}$ and that, for some integer $n_0 \geq 1$ and every compact $T \subset \mathcal{T}$,

$$(1)\quad \sup_{s\in T}\int_{\mathbb{R}^d} |M_s(u)|^{n_0}\,\mathrm{d}u < \infty, \qquad \sup\{|M_s(u)| : s \in T,\ |u| \geq \delta\} < 1 \quad \text{for every } \delta > 0.$$

With $n_0 = 1$ the first condition is that of Daniels (1954) and implies that X has a bounded continuous density; in general it implies this for the sum of n_0 observations rather than for X itself, and the power n_0 admits the chi-squared distributions of Section 5 with one degree of freedom, whose density is unbounded, and with two, whose density has a jump at the origin. Neither characteristic function is integrable. With $n_0 = 3$ the condition also admits the pairs of quadratic forms in three normal variables of Section 5.4, whose characteristic function decays like $|u|^{-1}$; the uniformity in s is what the proofs use. Both conditions exclude lattice distributions. They pass to the marginals of X, which the reduction of Section 3.3 requires; for $A \subseteq \{1,\dots,d\}$ nonempty let $K_A(s_A) = K(s)$ with $s_j = 0$ for $j \notin A$ be the cumulant generating function of X_A.

LEMMA 1. *If K satisfies (1) with the integer n_0, then K_A is finite on an open set containing the origin, has positive definite Hessian there, and satisfies (1) with n_0 replaced by $n_0 + d - |A|$.*

Throughout, ϕ and Φ are the standard normal density and distribution function, $\bar\Phi = 1 - \Phi$, and $\phi(\cdot, \Sigma)$ is the $N(0, \Sigma)$ density. For a d-vector ν of nonnegative integers write $|\nu| = \sum_{i=1}^d \nu_i$, $\nu! = \prod_{i=1}^d \nu_i!$, $t^\nu = \prod_{i=1}^d t_i^{\nu_i}$, $\lambda \leq \nu$ for componentwise inequality, and e_i for the i-th unit vector; a sum over $|\nu| = k$ runs over the finitely many such ν with $|\nu| = k$. Let $\hat s$ be the saddlepoint, $K'(\hat s) = \bar x$, and

$$(2)\qquad \hat K = K(\hat s), \qquad \hat K'' = K''(\hat s), \qquad \hat K_\nu = \frac{\partial^{|\nu|} K(\hat s)}{\partial s_1^{\nu_1}\cdots\partial s_d^{\nu_d}}, \qquad \kappa_\nu = \frac{\hat K_\nu}{\nu!}.$$

The Taylor expansion of K about $\hat s$ is

$$(3)\qquad K(\hat s + t) = \hat K + \bar x^\top t + \frac{t^\top \hat K'' t}{2} + G(t), \qquad G(t) = \sum_{k=3}^\infty \sum_{|\nu|=k} \kappa_\nu t^\nu,$$

and we define $c_{r,\nu}$ by

$$(4)\qquad \frac{G(t)^r}{r!} = \sum_{k=3r}^\infty \sum_{|\nu|=k} c_{r,\nu} t^\nu, \qquad \exp(G(t)) = 1 + \sum_{r=1}^\infty \sum_{k=3r}^\infty \sum_{|\nu|=k} c_{r,\nu} t^\nu.$$

Then $c_{1,\nu} = \kappa_\nu$ and, since $G^r/r! = (G/r)\cdot G^{r-1}/(r-1)!$,

$$(5)\qquad c_{r,\nu} = \frac{1}{r}\sum_{k=3}^{|\nu|-3(r-1)} \sum_{\substack{|\lambda|=k\\ \lambda\leq\nu}} \kappa_\lambda\, c_{r-1,\nu-\lambda}, \qquad r \geq 2.$$

In the terminology of Withers and Nadarajah (2008), $r!\,c_{r,\nu}$ is the ordinary multivariate partial Bell polynomial of multi-index ν and degree r in the κ_λ, and the recurrence (5) is a special case of that in Withers and Nadarajah (2010b, equation (8)). The coefficients $c_{r,\nu}$ are all that is needed to write every expansion in this paper. Every saddlepoint expansion below is obtained by expanding $\exp\{nG(t)\}$ along a contour on which t is of order $n^{-1/2}$ and integrating term by term: the term with $c_{r,\nu}$ carries the factor $n^{r-|\nu|/2} = n^{-m/2}$ with

$$m = |\nu| - 2r, \tag{6}$$

and $r \le m$ because $|\nu| \ge 3r$, so that the terms of relative order $n^{-m/2}$ are the finitely many with $|\nu| = m + 2r$, $r = 1, \dots, m$, and the expansions differ only in the normal integrals that the coefficients multiply. The κ_ν are not standardized; the matrix $\hat{K}''$ enters instead through the Hermite polynomials $\mathrm{He}_\nu(0, \hat{K}'')$ of Theorem 3 and the integrals $I(\nu)$ of Theorem 1, and the standardized univariate version of the recursion is given in Section 4. The storage of multi-indexed arrays is discussed in Appendix C.

The normal integrals are expressed through the multivariate Hermite polynomials

$$\mathrm{He}_\nu(x, \Sigma) = \frac{(-1)^{|\nu|}}{\phi(x, \Sigma)} \frac{\partial^{|\nu|} \phi(x, \Sigma)}{\partial x^\nu}. \tag{7}$$

The value at $x = 0$ is $\mathrm{He}_\nu(0, \Sigma) = \mathrm{i}^{|\nu|} E(S^\nu)$, where the random vector $S = (S_1, \dots, S_d)^\top$ is $N(0, \Sigma^{-1})$ and $S^\nu = \prod_{i=1}^d S_i^{\nu_i}$; it vanishes for $|\nu|$ odd and satisfies the recursion

$$\mathrm{He}_{\nu + e_k}(0, \Sigma) = -\sum_{j=1}^d \sigma^{jk} \nu_j \mathrm{He}_{\nu - e_j}(0, \Sigma), \qquad \mathrm{He}_0(0, \Sigma) = 1, \tag{8}$$

where σ^{jk} is the (j, k) element of Σ^{-1} and the terms with $\nu_j = 0$ are zero. This is the case $x = 0$ of the recursion $\mathrm{He}_{\nu + e_k}(x, \Sigma) = (\Sigma^{-1} x)_k \mathrm{He}_\nu(x, \Sigma) - \sum_{j=1}^d \sigma^{jk} \nu_j \mathrm{He}_{\nu - e_j}(x, \Sigma)$, which follows from equations (A.7) and (A.12) of Takemura and Takeuchi (1988). In one dimension $\mathrm{He}_m(x, 1) = \mathrm{He}_m(x)$ is the Hermite polynomial, $(-1)^m \phi^{(m)} = \mathrm{He}_m \phi$, and $\mathrm{He}_{2i}(0) = h_i$, $\mathrm{He}_{2i+1}(0) = 0$, with $h_i = (-2)^i (1/2)_i = \prod_{j=1}^i (1 - 2j)$.

3. The multivariate distribution function and density.

3.1. *The distribution function expansion.* Consider the orthant probability $\mathbb{P}(\bar{X} \ge \bar{x})$, and suppose first that the saddlepoint is in the positive orthant, $\hat{s} > 0$. The inversion formula (Kolassa, 2003, Section 2) is

$$\mathbb{P}(\bar{X} \ge \bar{x}) = \frac{1}{(2\pi \mathrm{i})^d} \int_{\hat{s} - \mathrm{i}\infty}^{\hat{s} + \mathrm{i}\infty} \exp\{n(K(s) - s^\top \bar{x})\} \prod_{j=1}^d \frac{\mathrm{d}s_j}{s_j}, \tag{9}$$

where the contour $\mathrm{Re}\, s = \hat{s}$ must lie to the right of the poles at $s_j = 0$. Expanding the exponent about $\hat{s}$ as in (3) and writing

$$\tfrac{1}{2}(s - \hat{s})^\top \hat{K}''(s - \hat{s}) = \tfrac{1}{2} s^\top \hat{K}'' s - s^\top \hat{K}'' \hat{s} + \tfrac{1}{2} \hat{s}^\top \hat{K}'' \hat{s}, \tag{10}$$

the integrand becomes a Gaussian factor times $\exp\{nG(s - \hat{s})\}$, and (4) gives the expansion term by term in the integrals of Kolassa (2003),

$$I(\Sigma, r, m, t, \hat{s}) = \frac{1}{(2\pi \mathrm{i})^d} \int_{\hat{s} - \mathrm{i}\infty}^{\hat{s} + \mathrm{i}\infty} \exp\left(\frac{s^\top \Sigma s}{2} - s^\top t\right) s^r (s - \hat{s})^m \, \mathrm{d}s, \tag{11}$$

for $r \ge -1$ and $m \ge 0$.

THEOREM 1. *Let $\hat{s} \geq 0$ componentwise, let $I(\nu) \equiv I(n\hat{K}'', -1, \nu, n\hat{K}''\hat{s}, \hat{s})$, where for $\hat{s}$ on the boundary of the orthant $I(\nu)$ is given by (14) of Section 3.2, which is continuous in $\hat{s}$, and let $E_n = \exp\{n(\hat{K} - \hat{s}^\top \bar{x} + \hat{s}^\top \hat{K}''\hat{s}/2)\}$. Then*

$$\mathbb{P}(\bar{X} \geq \bar{x}) = E_n \left[I(0) + \sum_{m=1}^{p} \sum_{r=1}^{m} \sum_{|\nu|=m+2r} n^r c_{r,\nu} I(\nu) \right] \{1 + O(n^{-(p+1)/2})\}, \tag{12}$$

with the error uniform for $\hat{s}$ in compact subsets of the intersection of $\mathcal{T}$ with the closed positive orthant.

Here $I(0) = \mathbb{P}(Z \geq \sqrt{n}\hat{K}''\hat{s})$, $Z \sim N(0, \hat{K}'')$, is the multivariate normal orthant probability, and $E_n I(0)$ is the leading term of Kolassa (2003). With $q = \hat{s}^\top \hat{K}''\hat{s}$, the proof (Appendix B) gives, uniformly for $\hat{s}$ in compact subsets of the intersection of $\mathcal{T}$ with the closed positive orthant,

$$|I(\nu)| \leq C_\nu e^{-nq/2} n^{-|\nu|/2} \prod_{j=1}^{d} \frac{1}{1 + \sqrt{n}\hat{s}_j}, \qquad C^{-1} \leq I(0)\, e^{nq/2} \prod_{j=1}^{d} (1 + \sqrt{n}\hat{s}_j) \leq C,$$

so $I(\nu)/I(0) = O(n^{-|\nu|/2})$ (compare Kolassa, 2006, Theorem 6.11.3) and the term with (r, ν) is of relative order $n^{-m/2}$, as (6) says. The leading term $E_n I(0)$ is thus of exact order $e^{n(\hat{K} - \hat{s}^\top \bar{x})} \prod_{j=1}^{d} (1 + \sqrt{n}\hat{s}_j)^{-1}$: in the interior of the orthant $e^{n(\hat{K} - \hat{s}^\top \bar{x})} n^{-d/2} \prod_{j=1}^{d} \hat{s}_j^{-1}$, with $I(0) \sim e^{-nq/2} n^{-d/2} (2\pi)^{-d/2} |\hat{K}''|^{-1/2} \prod_{j=1}^{d} \hat{s}_j^{-1}$, and at a saddlepoint with k vanishing components $e^{n(\hat{K} - \hat{s}^\top \bar{x})} n^{-(d-k)/2}$. Unlike the density expansion of Theorem 3 below, the odd m do not drop out; the terms with $m \leq p$ give relative error $O(n^{-(p+1)/2})$, and we call the approximation with $p = 2M - 1$ the approximation of order M, with relative error $O(n^{-M})$. The terms with $m \leq 2$ are

$$n \sum_{|\nu|=3} \kappa_\nu I(\nu) + n \sum_{|\nu|=4} \kappa_\nu I(\nu) + n^2 \sum_{|\nu|=6} c_{2,\nu} I(\nu), \tag{13}$$

the three terms displayed in equation (4) of Kolassa (2003), whose approximation keeps only the first. The terms with $m \leq p$ need κ_ν with $|\nu| \leq p + 2$ and $I(\nu)$ with $|\nu| \leq 3p$.

The expansion is of Hauschildt–Robinson type: a normal orthant probability plus normal density terms, all multiplied by the common factor E_n, and its one-dimensional case is the expansion of Hauschildt (1969) and Robinson (1982) (Corollary 2); it has no singular terms. The alternative, of Lugannani–Rice type, which removes the poles of (9) by subtracting a function with the same poles whose integral is known, has in several dimensions no natural counterpart, since the normal orthant probability has no simple expression; the constructions of Wang (1990) and Kolassa and Li (2010) remove the poles one coordinate at a time, at the cost of the dependence on the order of the coordinates and of the removable singularities noted in Section 1. The price of the Hauschildt–Robinson form is that the normal orthant probability $I(0)$ and its derivatives have to be evaluated, and that saddlepoints outside the positive orthant need the reduction of Section 3.3.

It is instructive to compare the error bound in Theorem 1 with that for $p = 1$ in Kolassa (2003). His remainder is bounded by $e^{n(\hat{K} - \hat{s}^\top \bar{x})} O(n^{-1})$ in absolute terms, and he notes (Kolassa, 2003, p. 279) that this is uniformly exponentially small but not both relative and uniform, a uniform relative bound for tail probabilities being harder to obtain than for densities even in one dimension. Measured against the leading term, which is of exact order $e^{n(\hat{K} - \hat{s}^\top \bar{x})} \prod_{j=1}^{d} (1 + \sqrt{n}\hat{s}_j)^{-1}$ by the bounds above, his bound is $O\{n^{-1} \prod_{j=1}^{d} (1 + \sqrt{n}\hat{s}_j)\}$: it is relative, of order n^{-1}, in the moderate-deviation region $\hat{s} = O(n^{-1/2})$ for which

Hauschildt–Robinson-type approximations are designed, and it degrades to $O(n^{d/2-1})$ at a fixed saddlepoint. The bound of Theorem 1 is relative in both regions, of the full order $n^{-(p+1)/2}$, and it is uniform up to and including saddlepoints with vanishing components, where the leading term is no longer of order $n^{-d/2}$. The proof (Appendix B) writes the orthant probability as the Laplace transform over the orthant of the density of the mean under the distribution tilted by $\hat{s}$, the tilting representation of Esscher (1932), as used by Robinson (1982) and Daniels (1987, Section 3), and bounds the Edgeworth remainder of that density with a polynomial weight, so that no inverse power of a component of $\hat{s}$ appears: Lemma 4 is the product-weighted version, uniform in the tilt, of the nonuniform Edgeworth bound for densities (Bhattacharya and Ranga Rao, 1976, Theorems 19.2 and 22.1), formula (1.5.16) of Jensen (1995), and the density-side argument is that of Jensen's proof of his Proposition 2.3.1 for $\lambda < 1$, who also explains (p. 33) why the contour argument of Kolassa (2003) stops at an absolute bound in several dimensions. The difference lies in where the factor $\prod_{j=1}^{d}(1+\sqrt{n}\hat{s}_j)^{-1}$ of the leading term is recovered. On the contour the pole factors $\prod_{j=1}^{d}|\hat{s}_j + \mathrm{i}y_j/\sqrt{n}|^{-1}$ can only be bounded by inverse powers of the $\hat{s}_j$, so the error is measured at the scale of the exponential factor and a relative bound is lost as a component of $\hat{s}$ tends to zero; in the Laplace transform over the orthant the factor arises from integrating $e^{-\hat{u}^\top v}$ against a remainder bounded by $\prod_{j=1}^{d}(1+v_j^2)^{-1}$, whose transform $\prod_{j=1}^{d}\min(\pi/2, 1/\hat{u}_j)$ is of the same order as the leading term whether $\hat{u}_j = \sqrt{n}\hat{s}_j$ is large or zero. In one dimension the result is the uniform validity of the Hauschildt–Robinson approximation up to and including the mean, formula (2.2.6) with Proposition 2.3.1 of Jensen (1995), see also Daniels (1987) and Kolassa (2007); the Lugannani–Rice form is discussed in Remark 1.

3.2. *The integrals $I(\nu)$.* Since $(s_j - \hat{s}_j)s^r = s^{r+e_j} - \hat{s}_j s^r$, the binomial expansion of $(s-\hat{s})^m$ gives

$$I(\Sigma, -1, m, t, \hat{s}) = \sum_{\nu_1=0}^{m_1} \cdots \sum_{\nu_d=0}^{m_d} \binom{m}{\nu} (-\hat{s})^\nu J(m-\nu-1), \qquad J(r) \equiv I(\Sigma, r, 0, t, \hat{s}), \tag{14}$$

and $J(r) = \prod_{j=1}^{d}(-D_j)^{r_j+1}\bar{\Phi}(t,\Sigma)$, where $\binom{m}{\nu} = \prod_{i=1}^{d}\binom{m_i}{\nu_i}$, $D_j = \partial/\partial t_j$, and $\bar{\Phi}(t,\Sigma) = \mathbb{P}(Z \geq t)$, $Z \sim N(0,\Sigma)$ (Kolassa, 2003, equations (6)–(7)). Thus all the integrals reduce to the derivatives $J(r)$, $r \geq -1$, of the multivariate normal orthant probability. For $r \geq 0$, $J(r) = \phi(t,\Sigma)\mathrm{He}_r(t,\Sigma)$. In general let $\mathcal{I} = \{i : r_i = -1\}$ and $\mathcal{J} = \{j : r_j \geq 0\}$, and write $t_\mathcal{J}$ and $r_\mathcal{J}$ for the subvectors of t and r with the components in $\mathcal{J}$ and $\Sigma_{\mathcal{I}\mathcal{J}}$ for the submatrix of Σ with rows in $\mathcal{I}$ and columns in $\mathcal{J}$, and similarly for the other index sets. Differentiating the orthant probability with respect to the components in $\mathcal{J}$ gives the marginal density of $Z_\mathcal{J}$ times the conditional orthant probability of $Z_\mathcal{I}$, so

$$J(r) = \left(\prod_{j\in\mathcal{J}}(-D_j)^{r_j}\right) g(t), \qquad g(t) = \phi(t_\mathcal{J}, \Sigma_{\mathcal{J}\mathcal{J}})\bar{\Phi}(t_\mathcal{I} - \Sigma_{\mathcal{I}\mathcal{J}}\Sigma_{\mathcal{J}\mathcal{J}}^{-1}t_\mathcal{J}, \Sigma_{\mathcal{I}\mathcal{I}\cdot\mathcal{J}}), \tag{15}$$

with $\Sigma_{\mathcal{I}\mathcal{I}\cdot\mathcal{J}}$ the conditional covariance matrix $\Sigma_{\mathcal{I}\mathcal{I}} - \Sigma_{\mathcal{I}\mathcal{J}}\Sigma_{\mathcal{J}\mathcal{J}}^{-1}\Sigma_{\mathcal{J}\mathcal{I}}$. When all components of r are -1 or 0, that is, when $r_\mathcal{J} = 0$, (15) reduces to $J(r) = g(t)$, which is equation (8) of Kolassa (2003); for general r it expresses $J(r)$ as a derivative of the same function g. The derivatives are generated by the following recursion, proved in Appendix B.

LEMMA 2. *For $j \in \mathcal{J}$, with $(\cdot)_{(j)}$ denoting the entry corresponding to j within $\mathcal{J}$,*

$$J(r+e_j) = (\Sigma_{\mathcal{J}\mathcal{J}}^{-1} t_{\mathcal{J}})_{(j)} J(r) - \sum_{\substack{k \in \mathcal{J} \\ r_k > 0}} r_k (\Sigma_{\mathcal{J}\mathcal{J}}^{-1})_{(j,k)} J(r-e_k)$$

$$- \sum_{i \in \mathcal{I}} (\Sigma_{\mathcal{I}\mathcal{J}} \Sigma_{\mathcal{J}\mathcal{J}}^{-1})_{(i,j)} J(r+e_i). \tag{16}$$

When $\mathcal{I} = \emptyset$, (16) is ϕ times the recursion of Takemura and Takeuchi (1988) for the Hermite polynomials. Since $J(r+e_i)$, $i \in \mathcal{I}$, has the index set $\mathcal{I} \setminus \{i\}$, the $J(r)$ are computed in increasing order of $|\mathcal{I}|$, starting for each $\mathcal{I}$ from the boundary value (15) with $r_{\mathcal{J}} = 0$ and raising the components in $\mathcal{J}$ one at a time. One normal orthant probability of dimension $|\mathcal{I}|$ is needed for each subset $\mathcal{I}$, $2^d - 1$ in all. In two and three dimensions these reduce to integrals of dimension one and two, which quadrature evaluates to high accuracy (Appendix C); for $d \geq 4$ deterministic quadrature becomes impractical, and a Monte Carlo or quasi-Monte Carlo method such as that of Genz (1992) and Genz and Bretz (2009) is generally needed. Kolassa (2003) gives (16) for $|r+1| \leq 3$, his equations (9)–(11); the higher-order terms require the general form.

3.3. *Saddlepoints of mixed signs and corner probabilities.* The requirement $\hat{s} \geq 0$ in Theorem 1 is not equivalent to $\bar{x} \geq E(X)$ when $d \geq 2$: in the example of Section 5.5, the point $\bar{x} = E(X) + (1.0, 3.0)$ standard deviations has a saddlepoint with a negative first component, because a large second component is produced more readily by the common part of the two variables. For a general point, let $N = \{j : \hat{s}_j < 0\}$ and $N^c = \{1, \ldots, d\} \setminus N$. Since $\{\bar{X}_N \geq \bar{x}_N\}$ is the complement of $\bigcup_{j \in N} \{\bar{X}_j < \bar{x}_j\}$,

$$\mathbb{P}(\bar{X} \geq \bar{x}) = \sum_{T \subseteq N} (-1)^{|T|} \, \mathbb{P}(\tilde{X}_{N^c \cup T} \geq \tilde{x}_{N^c \cup T}), \tag{17}$$

where the tilde denotes negation of the components in T and a probability with an empty index set is 1. The identity holds for any set N of indices, with $\leq$ in place of $<$ in the terms because $\bar{X}$ has a density for $n \geq n_0$. The term with $T = N$ is a full-dimensional orthant probability of a vector with cumulant generating function $K(\epsilon_1 s_1, \ldots, \epsilon_d s_d)$, with $\epsilon_j = -1$ for $j \in N$ and $\epsilon_j = 1$ otherwise, whose saddlepoint $(\epsilon_1 \hat{s}_1, \ldots, \epsilon_d \hat{s}_d)$ has no negative component, so that Theorem 1 applies to it, including when $\hat{s}_j = 0$ for some $j \in N^c$, where the integrals $I(\nu)$ are evaluated through their orthant-integral form (14) (Appendix C); negation preserves (1), and the marginals satisfy it by Lemma 1. The terms with $T \subsetneq N$ are orthant probabilities of lower-dimensional marginals, whose saddlepoints must be recomputed and may again have mixed signs, in which case (17) is applied recursively. Kolassa (2003, Section 3) gives the same reduction as a recursion that negates one negative component at a time; (17) is its closed form. Lower and mixed corner probabilities are handled by negating the components with the $\leq$ sign. Equation (17) results from moving the contour in (9) across the poles $s_j = 0$, $j \in N$, and collecting residues, and it is the multivariate version of the reflection of the univariate expansion at the mean (Corollary 2). The absolute error of the sum is of the order of the error of its largest term, a property shared by all Hauschildt–Robinson-type approximations (Robinson, 1982; Kolassa, 2003); if the terms P_T of (17) have relative errors at most ε, the relative error of the sum is at most $\varepsilon \sum_{T \subseteq N} P_T / \mathbb{P}(\bar{X} \geq \bar{x})$, which is large when the terms nearly cancel. This ratio should be monitored; it is moderate in the examples of Section 5. Theorem 2 gives conditions on K and $\bar{x}$ under which the ratio stays bounded, so that the relative error bound of Theorem 1 survives the reduction, and under which it tends to one. For $T \subseteq N$ the term P_T is the orthant probability of the vector $X^{(T)}$

with components $\epsilon_j X_j$, $j \in N^c \cup T$, where $\epsilon_j = -1$ for $j \in T$ and $\epsilon_j = 1$ for $j \in N^c$, at the point $(\epsilon_j \bar{x}_j)_{j \in N^c \cup T}$; its cumulant generating function is K restricted to the coordinate subspace $\mathcal{S}_T = \{s : s_j = 0 \text{ for } j \in N \setminus T\}$, with the signs of the components in T reversed, and the saddlepoint equation of P_T reads, in the coordinates of K,

$$s^{(T)} \in \mathcal{S}_T \cap \mathcal{T}, \qquad K'_j(s^{(T)}) = \bar{x}_j \quad \text{for } j \in N^c \cup T, \tag{18}$$

the saddlepoint of $X^{(T)}$ being $(\epsilon_j s_j^{(T)})_{j \in N^c \cup T}$; for $T = N$, $s^{(N)} = \hat{s}$, and $s^{(\emptyset)} = 0$ when $N^c = \emptyset$. Let $\hat{P}_T$ be the approximation of order M of P_T given by Theorem 1 for $X^{(T)}$, with $\hat{P}_\emptyset = P_\emptyset = 1$ when $N^c = \emptyset$, and let

$$\hat{P}(\bar{x}) = \sum_{T \subseteq N} (-1)^{|T|} \hat{P}_T. \tag{19}$$

THEOREM 2. *Let $\hat{s} \in \mathcal{T}$ be the saddlepoint of $\bar{x}$, let N be a nonempty set of indices, with P_T, $s^{(T)}$, $\hat{P}_T$ and $\hat{P}$ defined by (17)–(19) for this N, and suppose that (a) for every $T \subseteq N$ with $N^c \cup T \neq \emptyset$, (18) has a solution with $s_j^{(T)} \geq 0$ for $j \in N^c$ and $s_j^{(T)} \leq 0$ for $j \in T$. Then there is C such that $\sum_{T \subseteq N} P_T \leq C \mathbb{P}(\bar{X} \geq \bar{x})$ for all n large enough, and*

$$\mathbb{P}(\bar{X} \geq \bar{x}) = \hat{P}(\bar{x})\{1 + O(n^{-M})\}, \tag{20}$$

both uniformly over compact sets of $\bar{x}$ on which (a) holds. If moreover (b) $K'_j(s^{(\emptyset)}) > \bar{x}_j$ for every $j \in N$, then there is $c > 0$ such that $P_T \leq P_\emptyset - \mathbb{P}(\bar{X} \geq \bar{x}) \leq e^{-cn} P_\emptyset$ for every nonempty $T \subseteq N$ and all n large enough, uniformly over compact sets on which (a) and (b) hold. When $N = \{j\}$ and $\hat{s}_j < 0$, (b) follows from (a).

Condition (a) says that every term of (17) has a saddlepoint in the closed positive orthant, so that Theorem 1 applies to it, with the regularity condition for the marginals involved supplied by Lemma 1, and (17) need not be applied recursively. For $T = N$, where $s^{(N)} = \hat{s}$, it says that N contains every j with $\hat{s}_j < 0$ and no j with $\hat{s}_j > 0$, so that N is the set of negative components of $\hat{s}$ up to indices with $\hat{s}_j = 0$, which is what allows the compact sets of the theorem to reach a boundary $\hat{s}_j = 0$, $j \in N$. Condition (b) says that at the saddlepoint of the marginal of X_{N^c} the tilted means of the components in N exceed their thresholds, so that the constraints $\bar{X}_N \geq \bar{x}_N$ are asymptotically inactive; equivalently, $s^{(\emptyset)}$ minimizes $K(s) - s^\top \bar{x}$ over the closed positive orthant, with strictly positive multipliers for the constraints in N. Under (a) alone these multipliers are nonnegative, and (b) is equivalent to $s_j^{(\{j\})} < 0$ for every $j \in N$, the strict form of (a) for the singletons (Appendix B); so (b) excludes only the degenerate case in which the saddlepoint of a term with a single negated component has that component equal to zero. Under (a) and (b) the terms with $T \neq \emptyset$ are exponentially smaller than the marginal term $P_\emptyset$, so that the ratio $\sum_{T \subseteq N} P_T / \mathbb{P}(\bar{X} \geq \bar{x})$ tends to one, although at the sample sizes of Section 5 the terms are of comparable size. Under (a) and (b) the marginal term $\hat{P}_\emptyset$ alone therefore has relative error $O(n^{-M})$ as well. What the alternating sum adds is uniformity: the bound (20) requires only (a) and holds uniformly on compact sets on which (a) holds, including points at which (b) fails; at such a point the term $P_{\{j\}}$ with $s_j^{(\{j\})} = 0$ has the saddlepoint $s^{(\emptyset)}$ and is of the same exact order as $P_\emptyset$, so that the relative error of $\hat{P}_\emptyset$ alone does not tend to zero. The constant c of the exponential comparison, on the other hand, degrades as a gap $K'_j(s^{(\emptyset)}) - \bar{x}_j$ tends to zero. When $N = \{j\}$ this happens exactly as $\hat{s}_j \to 0$, since then $s^{(\emptyset)} \to \hat{s}$, but when N has several elements a component of $\hat{s}$ tending to zero does not by itself make a gap vanish, the gaps being evaluated at $s^{(\emptyset)}$ and not at $\hat{s}$. Across a boundary $\hat{s}_j = 0$ at which N gains the element j the approximation (19) is continuous in $\bar{x}$

when (a) holds on both sides, for N where $\hat{s}_j \geq 0$ and for $N \cup \{j\}$ where $\hat{s}_j \leq 0$, the terms for $N \cup \{j\}$ being the j-marginals and j-reflections of those for N and agreeing with them term by term at every order (Appendix B), and (20), with N on the one closed side and $N \cup \{j\}$ on the other, then holds uniformly on compact sets that straddle the boundary. This is automatic at the first sign change, where the other components of $\hat{s}$ are positive, but at a later change it requires the j-th components of the saddlepoints of the terms to vanish together with $\hat{s}_j$. These are additional conditions, which need not hold. They hold along the whole boundary when the zero sets of these components coincide with that of $\hat{s}_j$, as when X_j is independent of the other components, or is symmetric about a point conditionally on them, so that all of them vanish exactly where $\bar{x}_j$ equals that point; without such structure the zero sets are different hypersurfaces, which meet only in a lower-dimensional set, and (a) fails on at least one side at the remaining points of the boundary. The recursive application of (17), which expands every term whose saddlepoint has a negative component, is nevertheless continuous in $\bar{x}$ wherever the saddlepoints it requires exist, since its value does not depend on which index set containing the negative components of $\hat{s}$ is used in (17) (Appendix B); what it lacks where (a) fails is the error bound. For the case in which (a) fails, the proof of Theorem 2 in Appendix B gives more: if s^* minimizes $K(s) - s^\top \bar{x}$ over the closed positive orthant, $A = \{j : s^*_j > 0\}$ and $K'_j(s^*) > \bar{x}_j$ for $j \notin A$, then $\mathbb{P}(\bar{X} \geq \bar{x}) = \mathbb{P}(\bar{X}_A \geq \bar{x}_A)\{1 + O(e^{-cn})\}$ whatever the signs of $\hat{s}$, so that Theorem 1 applied to the marginal of X_A alone has relative error $O(n^{-M})$. This is the appropriate approximation there, where (17) would have to be applied again to its own terms, because the ratio of the sum of the terms to $\mathbb{P}(\bar{X} \geq \bar{x})$ need not then stay bounded: when every component of $\hat{s}$ is negative and $\bar{x}_1$ exceeds its mean, as can happen under negative dependence, for instance, the recursion produces terms that tend to one while $\mathbb{P}(\bar{X} \geq \bar{x})$ tends to zero.

The positive-orthant hypothesis cannot in general be dropped. If (12) is applied directly at a saddlepoint with a negative component, it approximates a contribution away from the tilted mean by a local normal expansion, and its relative error in general does not tend to zero as $n \to \infty$: up to polynomial factors, the ratio of the approximation to the true probability behaves like $e^{-\Delta}$ with Δ proportional to n, so that it tends to zero, the relative error tending to -1, or to infinity, exponentially, according to the sign of Δ. To see why, tilt the distribution of $\bar{X}$ by $n\hat{s}$: for any $\hat{s} \in \mathcal{T}$,

$$\mathbb{P}(\bar{X} \geq \bar{x}) = e^{n(\hat{K} - \hat{s}^\top \bar{x})} \int_0^\infty \cdots \int_0^\infty e^{-n\hat{s}^\top u} \hat{f}(\bar{x} + u) \, \mathrm{d}u_1 \cdots \mathrm{d}u_d, \tag{21}$$

where $\hat{f}$ is the density of the mean of n observations from the distribution with cumulant generating function $K(\hat{s} + t) - K(\hat{s})$, which has mean $\bar{x}$ and covariance matrix $\hat{K}''/n$. The expansion (12) is the Edgeworth expansion of $\hat{f}$ about its mean, integrated term by term against the exponential factor, and the Edgeworth expansion is accurate in relative terms only within $O(n^{-1/2})$ of $\bar{x}$. When $\hat{s} > 0$ the factor $e^{-n\hat{s}^\top u}$ decreases in every component on $u \geq 0$ and confines the integrand to that neighborhood. When $\hat{s}_j < 0$ it increases with u_j, and the integrand with $\hat{f}$ replaced by its normal approximation is largest at a value of u_j that does not depend on n: the integral depends on $\hat{f}$ at a fixed distance from its mean, in the central part of the distribution of $\bar{X}_j$, where the normal approximation has the wrong exponent. Proposition 1 makes this precise for $d = 2$. Let $\ell(s_1) = \min_{s_2}\{K(s_1, s_2) - s_1\bar{x}_1 - s_2\bar{x}_2\}$, the profile log-likelihood of s_1 in the exponential family of Section 2 divided by $-n$, with minimizer $\tilde{s}_2(s_1)$, so that $\ell(\hat{s}_1) = \hat{K} - \hat{s}^\top \bar{x}$, $\ell'(\hat{s}_1) = 0$ and $\ell''(\hat{s}_1) = \hat{K}''_{11} - \hat{K}''^2_{12}/\hat{K}''_{22}$, and let

$$\Delta = n\big\{\ell(0) - \ell(\hat{s}_1) - \tfrac{1}{2}\hat{s}_1^2 \ell''(\hat{s}_1)\big\}. \tag{22}$$

PROPOSITION 1. *Let $d = 2$, let $\hat{s}$ be the saddlepoint of $\bar{x}$ with $\hat{s}_1 < 0 \le \hat{s}_2$, let the minimum defining $\ell(s_1)$ be attained for every $s_1 \in [\hat{s}_1, 0]$, and suppose that $\tilde{s}_2(0) \ge 0$, so that condition (a) of Theorem 2 holds with $N = \{1\}$, and that $\hat{s}_2 + \hat{K}''_{12}\hat{s}_1/\hat{K}''_{22} > 0$. Then the leading term $E_n I(0)$ of (12) evaluated at $\hat{s}$ satisfies*

$$\lim_{n\to\infty} \frac{1}{n} \log \frac{E_n I(0)}{\mathbb{P}(\bar{X} \ge \bar{x})} = -\frac{\Delta}{n} = \frac{1}{6}\hat{s}_1^3 \ell'''(\xi) \quad \textit{for some } \xi \in (\hat{s}_1, 0), \tag{23}$$

and every term $E_n n^r c_{r,\nu} I(\nu)$ of (12) at $\hat{s}$ is bounded by a power of n times $E_n I(0)$.

The proof is in Appendix B: by Theorem 2 the exponent of the probability is $n\ell(0)$, that of the marginal $\mathbb{P}(\bar{X}_2 \ge \bar{x}_2)$, while the exponent of $E_n I(0)$ is n times the minimum over the closed positive orthant of the quadratic approximation of $K(s) - s^\top \bar{x}$ about $\hat{s}$, which under the last hypothesis is attained on the face $s_1 = 0$ and equals the quadratic Taylor approximation of ℓ about $\hat{s}_1$ evaluated at 0. The hypothesis $\tilde{s}_2(0) \ge 0$ makes $\bar{X}_2 \ge \bar{x}_2$ the active constraint; without it both probabilities can tend to one, as for a normal vector with $\bar{x}$ below its mean in both components and $\hat{s}$ of mixed signs, and the sign of $\hat{s}_1$ alone does not determine the active constraints. The case $\hat{s}_1 < 0 \le \tilde{s}_2(0)$ can occur with $\bar{x}_1$ above its marginal mean, as in the example of Section 5.5. At a fixed $\bar{x}$, Δ is proportional to n, of either sign, and vanishes only where $\ell'''(\xi) = 0$: identically when ℓ is quadratic, as for a normal vector or for a normal first component independent of the second, in which cases the direct evaluation has the right exponent, and otherwise at most on an exceptional set of $\bar{x}$. Since the correction terms are at most polynomially larger than the leading term, the direct application of (12) at any fixed order is at most a power of n times $e^{-\Delta}$ relative to the true probability: for $\Delta > 0$ its relative error tends to -1, and for $\Delta < 0$ it is off by at least $e^{-\Delta}$ up to polynomial factors unless its terms cancel. In the example of Kolassa and Li (2010) of Section 5.1, at the two points $y = (2.5, 3.5)$ and $(2.5, 4.0)$ considered in the Supplementary Material, the hypotheses hold, with $\hat{s}_1 = -0.103$ and -0.179, $\tilde{s}_2(0) = 0.429$ and 0.500 and $\hat{s}_2 + \hat{K}''_{12}\hat{s}_1/\hat{K}''_{22} = 0.433$ and 0.515, and $\Delta/n = 0.0004$ and 0.0027: the direct approximation is off by the factors $e^{-0.0004n}$ and $e^{-0.0027n}$ up to polynomial factors, which the Supplementary Material shows taking hold from $n = 80$ on at $y = (2.5, 4.0)$ and from $n = 320$ at $y = (2.5, 3.5)$, where Δ/n is seven times smaller; at $y = (2.2, 4.0)$, $\Delta/n = 0.0145$. In terms of (9), moving the contour from $\operatorname{Re} s_1 > 0$ to $\operatorname{Re} s_1 = \hat{s}_1 < 0$ crosses the pole at $s_1 = 0$, whose residue is the marginal probability, with saddlepoint $\{0, \tilde{s}_2(0)\}$, $\tilde{s}_2(0)$ being the saddlepoint of the marginal of $\bar{X}_2$; (17) expands this residue about its own saddlepoint, while the direct application of (12) in effect expands it about $\hat{s}$. The two agree when $\hat{s}_1 = 0$, so the approximation obtained from (17) is continuous across $\hat{s}_1 = 0$ (term by term, Appendix B), as the univariate approximation is at the mean (Section 4, where the difference between the expansion of the upper tail evaluated at $\hat{s} < 0$ and one minus the expansion of the lower tail is shown to be small only near the mean). The Supplementary Material follows the growth with n of the error of the direct application on an example of Kolassa and Li (2010).

3.4. *The density expansion.* The density of $\bar{X}$ is obtained from the same expansion of $\exp\{nG\}$, with the moments of the normal distribution in place of the integrals $I(\nu)$: on the contour through $\hat{s}$ parallel to the imaginary axes, the term with $c_{r,\nu}$ integrates against the normal density with covariance matrix $(n\hat{K}'')^{-1}$ to $n^{r-|\nu|/2} c_{r,\nu} \mathrm{He}_\nu(0, \hat{K}'')$. Every term of the expansion is therefore the product of a coefficient given by (5) and a value of a Hermite polynomial given by (8).

THEOREM 3. *The saddlepoint expansion of the density of $\bar{X}$ is*

$$f_{\bar{X}}(\bar{x}) = \Big(\frac{n}{2\pi}\Big)^{\frac{d}{2}} |\hat{K}''|^{-\frac{1}{2}} \exp\{n(\hat{K} - \hat{s}^\top \bar{x})\}$$
$$\times \left[1 + \sum_{m=1}^{M-1} \frac{1}{n^m} \sum_{r=1}^{2m} \sum_{|\nu|=2m+2r} c_{r,\nu} \mathrm{He}_\nu(0, \hat{K}'')\right] \{1 + O(n^{-M})\}, \tag{24}$$

with the error relative and uniform for $\hat{s}$ in compact subsets of $\mathcal{T}$.

The proof is in Appendix B, and covers Daniels' expansion (Corollary 1) as the case $d = 1$. Equation (24) is the tilted Edgeworth expansion evaluated at the mean of the tilted distribution, written formally to all orders by Kolassa (2006, Section 6.4) and, with coefficients given by Bell polynomials, by Withers and Nadarajah (2010a, Section 5); the recursion (5) is the recurrence of Withers and Nadarajah (2010b) for multivariate Bell polynomials, and (8) follows from Takemura and Takeuchi (1988). With the terms organized by the order in n, however, every term of every order is computed numerically by the two recursions alone, without computer algebra. The m-th term needs κ_ν for $|\nu| \le 2m + 2$ and $\mathrm{He}_\nu(0, \hat{K}'')$ for $|\nu| \le 6m$. For $m = 1$ the bracket is

$$1 + \frac{1}{n}\Big\{ \sum_{|\nu|=4} \kappa_\nu \mathrm{He}_\nu(0, \hat{K}'') + \sum_{|\nu|=6} c_{2,\nu} \mathrm{He}_\nu(0, \hat{K}'')\Big\}, \tag{25}$$

and little arithmetic is involved: for $d = 2$, the four κ_ν with $|\nu| = 3$ and the five with $|\nu| = 4$ are read off from the derivatives; (5) with $r = 2$ gives the seven $c_{2,\nu}$ with $|\nu| = 6$ as $\frac{1}{2}\sum_{|\lambda|=3,\, \lambda \le \nu} \kappa_\lambda \kappa_{\nu-\lambda}$, sixteen products in all; (8) gives the fifteen $\mathrm{He}_\nu(0, \hat{K}'')$ with $|\nu| = 2, 4, 6$, each from at most two values with $|\nu|$ smaller by two; and the braces in (25) contain a sum of $5 + 7$ products. The term with $m = 2$ is obtained in the same way from the κ_ν with $|\nu| = 6$, the $c_{r,\nu}$ with $|\nu| = 4 + 2r$, $r = 2, 3, 4$, and the $\mathrm{He}_\nu(0, \hat{K}'')$ with $|\nu| \le 12$, and so on for every m, with the same code. The cost grows with the number of multi-indices with $|\nu| \le 6m$, which is $\binom{6m+d}{d}$.

In the usual derivation of (25), with $G_k(t) = \sum_{|\nu|=k} \kappa_\nu t^\nu$ and $S \sim N(0, (\hat{K}'')^{-1})$, the identity $\mathrm{He}_\nu(0, \hat{K}'') = \mathrm{i}^{|\nu|} E(S^\nu)$ shows that the term in braces in (25) is

$$E\{G_4(S)\} - \tfrac{1}{2} E\{G_3(S)^2\}.$$

The tensor method writes $G_3(S)$ and $G_4(S)$ as sums over three and four indices with weights $1/6$ and $1/24$, and evaluates the expectations by pairing the indices. The 3 pairings of four indices give the same contraction of the fourth derivatives, and the 15 pairings of six indices have to be sorted into the 9 that pair two indices within each third derivative and the 6 that pair every index across, with coefficients $3/24 = 1/8$, $-9/72 = -1/8$ and $-6/72 = -1/12$. The result is

$$1 + \frac{1}{n}\left(\frac{\bar{\rho}_4}{8} - \frac{\bar{\rho}_{13}^2}{8} - \frac{\bar{\rho}_{23}^2}{12}\right),$$

the familiar form of Barndorff-Nielsen and Cox (1979), McCullagh (1987, equation (5.10) and Section 6.3) and Kolassa (2006, equation (135)), with σ^{ij} the elements of $(\hat{K}'')^{-1}$ and the invariants

$$\bar{\rho}_4 = \sum_{i=1}^{d}\sum_{j=1}^{d}\sum_{k=1}^{d}\sum_{l=1}^{d} \hat{K}_{e_i+e_j+e_k+e_l}\, \sigma^{ij}\sigma^{kl},$$

$$\bar\rho_{13}^2 = \sum_{i=1}^d\sum_{j=1}^d\sum_{k=1}^d\sum_{l=1}^d\sum_{m=1}^d\sum_{p=1}^d \hat K_{e_i+e_j+e_k}\hat K_{e_l+e_m+e_p}\,\sigma^{ij}\sigma^{kl}\sigma^{mp},$$

$$\bar\rho_{23}^2 = \sum_{i=1}^d\sum_{j=1}^d\sum_{k=1}^d\sum_{l=1}^d\sum_{m=1}^d\sum_{p=1}^d \hat K_{e_i+e_j+e_k}\hat K_{e_l+e_m+e_p}\,\sigma^{il}\sigma^{jm}\sigma^{kp}.$$

At order n^{-2} the same method requires the expectations of $G_6(S)$, $G_3(S)G_5(S)$, $G_4(S)^2$, $G_3(S)^2G_4(S)$ and $G_3(S)^4$, which involve 15, 105, 105, 945 and 10395 pairings, to be sorted into 1, 2, 3, 7 and 8 distinct invariants, 21 in all, each with its coefficient, and the enumeration grows accordingly at each further order. McCullagh (1987) and Kolassa (2006) stop at order n^{-1}, and the term of order n^{-2} does not seem to have been written out in terms of invariants. In (24) the term of any order is easy to compute, since the $c_{r,\nu}$ and the $\mathrm{He}_\nu(0,\hat K'')$ are obtained by the simple recursions (5) and (8).

4. One dimension. For $d=1$ the expansions of Section 3 give the classical expansions of the density and the distribution function of a mean (and, in Appendix A, of a ratio), all as corollaries, and it is convenient to write them with standardized coefficients. Let $K(s)=\log E(e^{sX})$, let $\hat s$ be the saddlepoint, $K'(\hat s)=\bar x$, and let $\hat K_k = K^{(k)}(\hat s)$. Define the standardized Taylor coefficients and the two basic quantities

$$\theta_k = \frac{\hat K_k}{k!\,\hat K_2^{k/2}}, \qquad \hat w = \mathrm{sgn}(\hat s)\sqrt{2(\hat s\bar x - K(\hat s))}, \qquad \hat u = \hat s\sqrt{\hat K_2}, \tag{26}$$

so that $\theta_2=1/2$; $\hat w$ is the signed root of the likelihood ratio statistic for $s=0$ in the exponential family of Section 2, and $\hat u$ the Wald statistic, the maximum likelihood estimate $\hat s$ standardized by the observed information, both per observation. Let $\Theta(s)=\sum_{k=1}^\infty \theta_{k+2}s^k$ be the generating function of the coefficients of degree three and higher, and define $d_{r,j}$ by

$$\sum_{j=0}^\infty d_{r,j}s^j = \frac{\Theta(s)^r}{r!}, \qquad r\ge 0. \tag{27}$$

Then $d_{0,0}=1$, $d_{0,j}=0$ for $j\ge 1$, $d_{1,j}=\theta_{j+2}$, $d_{r,j}=0$ for $j<r$, and

$$d_{r,j} = \frac{1}{r}\sum_{k=1}^{j-r+1}\theta_{k+2}\,d_{r-1,j-k}, \qquad r\ge 1,\ j\ge r, \tag{28}$$

which is (5) in one dimension: $\kappa_k=\theta_k\hat K_2^{k/2}$ and $c_{r,m+2r}=d_{r,m}\hat K_2^{(m+2r)/2}$, so that $d_{r,m}$ is the standardized coefficient of $(s-\hat s)^{m+2r}$ in the r-th power of the remainder of K, which appears at relative order $n^{-m/2}$. In the terminology of Comtet (1974, Section 3.3), $r!\,d_{r,j}$ is the ordinary partial Bell polynomial of degree r and weight j in $\theta_3,\theta_4,\ldots$, and (28) is the counterpart for these polynomials of his recurrence (3k) on p. 136 for the exponential ones. Let also

$$a_{i,j} = \sum_{k=0}^j d_{k,j}\,h_{i+k}, \qquad i,j\ge 0, \tag{29}$$

with $h_i=\mathrm{He}_{2i}(0)$ as in Section 2, so that $a_{i,0}=h_i$. Since $h_{i+k}=h_i(-2)^k(i+1/2)_k$ and $\sum_{k=0}^\infty (c)_k z^k/k! = (1-z)^{-c}$, the $a_{i,j}$ have the generating function

$$\sum_{j=0}^\infty a_{i,j}s^j = \sum_{k=0}^\infty h_{i+k}\frac{\Theta(s)^k}{k!} = h_i\{1+2\Theta(s)\}^{-i-1/2}, \tag{30}$$

as formal power series; as functions, the series converges on the largest disc $|s| < R$ on which Θ is analytic and $1 + 2\Theta$ has no zero, with the branch of the power equal to 1 at $s = 0$, so that the right side equals h_i there, and this disc can be smaller than the disc of convergence of Θ. Up to the factor h_i, the $a_{i,j}$ are the potential polynomials of Comtet (1974, Section 3.5, Theorem B) of $1 + 2\Theta$ with the exponent $-i - 1/2$, in ordinary rather than exponential form.

Theorem 3 with $d = 1$ gives Daniels' expansion of the density of $\bar{X}$.

COROLLARY 1 (Daniels). *The saddlepoint expansion of the density of $\bar{X}$ is*

$$f_{\bar{X}}(\bar{x}) = \frac{\sqrt{n}\,\phi(\sqrt{n}\hat{w})}{\sqrt{\hat{K}_2}} \left\{ \sum_{i=0}^{M-1} \frac{a_i}{n^i} + O(n^{-M}) \right\}, \qquad a_i = a_{i,2i} = \sum_{k=0}^{2i} d_{k,2i} h_{i+k}, \tag{31}$$

so that $a_0 = 1$ and $a_1 = 3\theta_4 - \frac{15}{2}\theta_3^2$, with the error relative and uniform on compact subsets of the range of K'.

The term of order n^{-1} and the uniformity of the error on compact sets are formula (2.2.4) and Proposition 2.3.1 of Jensen (1995). Indeed, for $d = 1$ the prefactor of (24) is $(n/2\pi\hat{K}_2)^{1/2} \exp\{n(K(\hat{s}) - \hat{s}\bar{x})\}$, Daniels' leading term, and since $\mathrm{He}_{2j}(0, \hat{K}_2) = h_j \hat{K}_2^{-j}$ the term with $c_{r,2m+2r}$ in (24) is $d_{r,2m} h_{m+r}$, whose sum over $r = 1, \ldots, 2m$ is a_m. The i-th coefficient is obtained from the $d_{r,2i}$ with $r \le 2i$, that is, from $\theta_3, \ldots, \theta_{2i+2}$, and the next coefficient is

$$a_2 = -15\theta_6 + 105\theta_3\theta_5 + \tfrac{105}{2}\theta_4^2 - \tfrac{945}{2}\theta_3^2\theta_4 + \tfrac{3465}{8}\theta_3^4.$$

For the distribution function, let $\hat{u}_n = \sqrt{n}\hat{u}$ and $\hat{w}_n = \sqrt{n}\hat{w}$, so that

$$\exp\{(\hat{u}_n^2 - \hat{w}_n^2)/2\} = \exp\{n(K(\hat{s}) - \hat{s}\bar{x} + \hat{s}^2\hat{K}_2/2)\} = E_n, \tag{32}$$

the factor of Theorem 1, and write the inversion integral (9) as

$$\mathbb{P}(\bar{X} > \bar{x}) = \frac{1}{2\pi\mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} e^{n(K(s) - s\bar{x})} \frac{\mathrm{d}s}{s}, \qquad c > 0. \tag{33}$$

The integrals $I(\nu)$ become, after the substitution $v = (n\hat{K}_2)^{1/2} s$, $I(\nu) = I_\nu(\hat{u}_n)/(n\hat{K}_2)^{\nu/2}$, where, for real u and $m \ge 0$,

$$\begin{aligned} I_m(u) &= e^{-u^2/2} \int_0^\infty e^{-ut} \mathrm{He}_m(t) \phi(t)\, \mathrm{d}t \\ &= \frac{1}{2\pi\mathrm{i}} \int_{u-\mathrm{i}\infty}^{u+\mathrm{i}\infty} e^{v^2/2 - vu} (v-u)^m \frac{\mathrm{d}v}{v} \qquad (u > 0). \end{aligned} \tag{34}$$

Then $I_0(u) = \Phi(-u)$ and, from $(v-u)^{m+1} = (v-u)^m (v-u)$ and

$$\frac{1}{2\pi\mathrm{i}} \int_{u-\mathrm{i}\infty}^{u+\mathrm{i}\infty} e^{v^2/2 - vu} (v-u)^m \,\mathrm{d}v = \mathrm{He}_m(0)\phi(u),$$

$$I_{m+1}(u) = \mathrm{He}_m(0)\phi(u) - uI_m(u), \qquad \mathrm{He}_{2r}(0) = h_r, \quad \mathrm{He}_{2r+1}(0) = 0, \tag{35}$$

which is Lemma 2 and (14) in one dimension. These are the integrals of Daniels (1987); see Kolassa (2003) and Kolassa (2006, Section 5.2) for the contour form and the recursion. Kolassa (2006, Section 5.2) states that the series may be truncated after any number of terms with an error of the order of the first term omitted; the proof in Kolassa (2003, Section 3) covers truncation after the first correction term. Since $n^r c_{r,m+2r} I(m+2r) = n^{-m/2} d_{r,m} I_{m+2r}(\hat{u}_n)$, Theorem 1 and the reflection of Section 3.3 give the following.

COROLLARY 2 (Hauschildt–Robinson). *Let $p \geq 0$ be an integer and let E_n be as in (32). For $\hat{s} \geq 0$,*

$$\mathbb{P}(\bar{X} > \bar{x}) = E_n \left[I_0(\hat{u}_n) + \sum_{m=1}^{p} \frac{1}{n^{m/2}} \sum_{r=1}^{m} d_{r,m} I_{m+2r}(\hat{u}_n) \right] \{1 + O(n^{-(p+1)/2})\}. \tag{36}$$

For $\hat{s} \leq 0$,

$$\mathbb{P}(\bar{X} < \bar{x}) = E_n \left[I_0(|\hat{u}_n|) + \sum_{m=1}^{p} \frac{(-1)^m}{n^{m/2}} \sum_{r=1}^{m} d_{r,m} I_{m+2r}(|\hat{u}_n|) \right] \{1 + O(n^{-(p+1)/2})\}. \tag{37}$$

For $p = 1$, (36) is the approximation of Hauschildt (1969), Robinson (1982) and Daniels (1987, equation (5.3)), and formula (2.2.6) of Jensen (1995), whose Proposition 2.3.1 gives the uniformity of its error for $\hat{s}$ in compact sets, including $\hat{s} = 0$, in the form of the bound stated after Theorem 1: with $I_3(u) = (u^2 - 1)\phi(u) - u^3\Phi(-u)$,

$$\mathbb{P}(\bar{X} > \bar{x}) \approx E_n \left[\Phi(-\hat{u}_n) \left(1 - \frac{\theta_3 \hat{u}_n^3}{\sqrt{n}} \right) + \frac{\theta_3(\hat{u}_n^2 - 1)\phi(\hat{u}_n)}{\sqrt{n}} \right]. \tag{38}$$

Equation (37) is (36) applied to $-\bar{X}$, whose saddlepoint is $-\hat{s}$ and whose θ_k are $(-1)^k\theta_k$, hence $d_{r,m} \to (-1)^m d_{r,m}$. The two formulas join smoothly at $\hat{s} = 0$. The function I_m is entire, and $\int_{-\infty}^{\infty} e^{-ut}\mathrm{He}_m(t)\phi(t)\,\mathrm{d}t = (-u)^m e^{u^2/2}$ gives $I_m(u) + (-1)^m I_m(-u) = (-u)^m$ for all real u. Substituting this into (36) evaluated at a negative $\hat{u}_n$ shows that (36) minus $1-$(37) equals

$$E_n \sum_{m=0}^{p} \sum_{r=0}^{m} d_{r,m} (n\hat{u}^2)^r |\hat{u}|^m - 1, \qquad d_{0,0} = 1; \tag{39}$$

since $\hat{w}^2 = \hat{u}^2\{1 + 2\Theta(-\hat{u})\}$ and $\hat{u} < 0$, $E_n = \exp\{-n\hat{u}^2\Theta(|\hat{u}|)\}$, while the untruncated double sum is $\exp\{n\hat{u}^2\Theta(|\hat{u}|)\} = E_n^{-1}$, so the difference is $O(n|\hat{u}|^{p+3})$ as $\hat{u} \to 0$ with $n\hat{u}^2$ bounded, in particular for fixed n. Hence the approximation with terms $m \leq p$, which is analytic on either side of the mean, is $p + 2$ times continuously differentiable at the mean, and no special treatment of the mean is needed; away from the mean, however, the difference is of exponential order in n, which is the one-dimensional case of the argument of Section 3.3.

At $\hat{s} = 0$, $\hat{u}_n = \hat{w}_n = 0$, and the identity above and (35) give $I_0(0) = \frac{1}{2}$, $I_{2j}(0) = 0$ for $j \geq 1$ and $I_{2j+1}(0) = h_j\phi(0)$ for $j \geq 0$. The terms with m even therefore vanish, those with $m = 2i + 1$ change sign between (36) and (37), and by (29)

$$\sum_{r=1}^{2i+1} d_{r,2i+1} I_{2i+1+2r}(0) = \phi(0) \sum_{r=1}^{2i+1} d_{r,2i+1} h_{i+r} = \phi(0)\, a_{i,2i+1},$$

since $d_{0,2i+1} = 0$. Both (36) and (37) thus give, at the mean, where both carry the error bound of Theorem 1,

$$\mathbb{P}(\bar{X} \leq \bar{x}) \approx \frac{1}{2} - \phi(0) \sum_{i=0}^{\lfloor (p-1)/2 \rfloor} \frac{a_{i,2i+1}}{n^{i+1/2}}. \tag{40}$$

The Supplementary Material applies Corollaries 1 and 2 with orders one to six to the non-central chi-squared distribution at 202,000 points: the median relative error falls from 10^{-3}–10^{-5} at order one to 10^{-13}–10^{-16} at order six, and with fewer than ten degrees of freedom the maximum error stops improving after order three or four.

REMARK 1 (The Lugannani–Rice expansion). The expansion of Lugannani and Rice (1980) removes the pole of (33) by subtracting a function with the same pole whose integral is a normal tail probability, and expands the remainder; it does not follow from Theorem 1, and its construction has no counterpart in several dimensions. In the present notation it reads, for $\bar{x} \neq E(X)$,

$$\mathbb{P}(\bar{X} \leq \bar{x}) = \Phi(\hat{w}_n) - \phi(\hat{w}_n) \sum_{i=0}^{M-1} \frac{b_i}{n^{i+1/2}} + O(n^{-M}), \tag{41}$$

where

$$b_i = \sum_{j=0}^{2i} \frac{(-1)^j a_{i,j}}{\hat{u}^{2i+1-j}} - \frac{h_i}{\hat{w}^{2i+1}},$$

and the error is relative to the leading term; for $M = 1$ it is the familiar $\Phi(\hat{w}_n) + \phi(\hat{w}_n)(\hat{w}_n^{-1} - \hat{u}_n^{-1})$. The two parts of b_i are individually singular at the mean, but by (30) the singularities cancel to all orders, with $b_i \to a_{i,2i+1}$, so that at the mean (41) with M terms equals (40) with $p = 2M - 1$; the proofs, the derivative of (41), which is Daniels' expansion with $M + 1$ terms up to a term of relative order n^{-M} (for $M = 1$ the adjusted approximation of Routledge and Tsao (1997)), and the series for b_i used near the mean are in the Supplementary Material (Broda and Kan, 2026). Kolassa (2007) proves that for $p = 1$ the approximations of Robinson and of Lugannani and Rice agree to relative order n^{-1} uniformly for $\hat{s} \geq 0$, and Jensen (1995, formula (3.3.9), from Theorem 3.2.1) gives the Lugannani–Rice formula with an error $\phi(\hat{w}_n)O(n^{-3/2})$ uniform across the mean; our numerical comparisons use the Hauschildt–Robinson form throughout.

5. Applications. We illustrate the gains from the higher-order terms in a number of applications. In Sections 5.1 to 5.3 the variable is the mean of n independent observations, and n is a sample size; the Supplementary Material also lets n grow with the point fixed, so that the rates $O(n^{-M})$ of Theorem 1 can be read off. In the remaining examples the "n" of the expansions is the degrees of freedom or the noncentrality rather than a sample size; the expansions are written with $n = 1$ and the cumulant generating function of the variable itself; the ratio of regression coefficients of Section 5.4 is the mean of ν independent pairs whose distribution depends on ν through the standard errors, and is within the setting when these are proportional to $\nu^{-1/2}$ with the coefficients fixed, as in the usual asymptotics of a regression. Throughout, the approximation of order M has the terms $m \leq 2M - 1$ of (12), with (17) where the saddlepoint has a negative component, and the relative error of a density approximation $\hat{f}$ is $(\hat{f} - f)/f$, that of an approximation $\hat{F}$ to a distribution function is $(\hat{F} - F)/\min(F, 1 - F)$, except in Table 1, where the errors are relative to F itself to match the values of Wang (1990), and that of an approximation $\hat{P}$ to an orthant probability is $(\hat{P} - P)/P$; the tables below show these errors with their signs. All computations were done in MATLAB in double precision, the multivariate expansions being evaluated in the whitened coordinates of Appendix C. This is not a refinement: the terms of a given order are not individually invariant under linear maps of X, only their sum is, and in the original coordinates they cancel at order five by factors of up to 5×10^9 at the points of Table 2, 5×10^8 at those of Table 1 and 2×10^{17} in the autocorrelation example of the Supplementary Material, so that double precision then loses the entries of the highest orders; in the whitened form the cancellation indicator of Appendix C stays below 1.2×10^2 in Sections 5.1 and 5.2 and below 5×10^4 throughout. A bivariate point takes a fraction of a second at any order, and a point of Table 4 at order five about a minute. The expansions of Table 6 were checked against an independent

implementation in 40-digit arithmetic (Section 5.5), and the exact values were obtained as described in each subsection. The integrals I_m of (34) were evaluated by quadrature when $|\hat{u}_n| > 2.5$, because the recursion (35) loses all accuracy there (Appendix C). The MATLAB programs that implement the expansions, together with the scripts and data that reproduce the tables of this section and of the Supplementary Material, and a user manual, are provided in the Supplementary Material (Broda and Kan, 2026) and at https://raymond-kan.github.io/research.html. Packages that implement the expansions for a user-supplied cumulant generating function, every derivative being generated by Taylor-mode automatic differentiation, are available for Julia (`Pkg.add("SaddlepointExpansions")`), Python (`pip install saddlepoint-expansions`) and MATLAB (https://github.com/raykan3/SaddlepointExpansions-matlab).

5.1. *The bivariate examples of Wang (1990) and Kolassa and Li (2010).* The first-order approximations of Wang (1990) and Kolassa and Li (2010) were discussed in Section 1. This subsection compares them with the expansions of Section 3 on the examples of their own papers, with exact values in place of the simulations used there, at the sample sizes of their tables (the Supplementary Material lets n grow with the point fixed); Section 5.2 takes up the Laplace example of Section 1, on which neither approximation is defined.

Wang's example. Let $X = U_0 + U_1 - 2$ and $Y = U_0 + U_2 - 2$ with U_0, U_1, U_2 independent unit exponential variables, a pair with a common component as in Section 5.5 below, and let $(\bar{X}, \bar{Y})$ be the mean of $n = 10$ pairs. Wang (1990, Table 1) gives $F(x, y) = \mathbb{P}(\bar{X} \le x, \bar{Y} \le y)$ at 45 points by simulation with 10^6 samples, described as reliable to the digits given, by his approximation, and by the normal approximation. The exact value is the one-dimensional integral

$$F(x,y) = \int_0^{\min(x,y)+2} g(u)\, G(x+2-u)\, G(y+2-u)\, \mathrm{d}u,$$

where g and G are the density and the distribution function of the gamma distribution with shape n and scale $1/n$. Our values are the expansions for $(-\bar{X}, -\bar{Y})$. Table 1 shows a selection of the points. Over the 40 points at which both of Wang's printed values have at least two significant digits, the largest relative errors are 4.3% for his simulation and 6.7% for his approximation (at least 4.8% after allowing for the rounding of the printed values), while those of the expansions of orders one to five are 1.7×10^{-2}, 4.2×10^{-4}, 1.2×10^{-5}, 1.1×10^{-6} and 7.6×10^{-8} over all 44 points with a positive printed value. His approximation is least accurate in the corners, where his discussion places the probabilities of most interest: relative to the small probability $1 - F$ it errs by 31% at $(2, 2)$, against 1.6×10^{-3} at order one.

Kolassa and Li's example. Let $Y_1 = X_1 + X_2$ and $Y_2 = X_2 + X_3$ with X_1, X_2, X_3 independent unit exponential variables, let $(\bar{Y}_1, \bar{Y}_2)$ be the mean of $n = 5$ pairs, and consider $\mathbb{P}(\bar{Y}_1 \ge y_1, \bar{Y}_2 \ge y_2) = \int_0^\infty g(u)\bar{G}(y_1 - u)\bar{G}(y_2 - u)\, \mathrm{d}u$, with $\bar{G} = 1 - G$ and g, G as above with shape 5 and scale $1/5$. Kolassa and Li (2010, Table 1) give their approximation, that of Kolassa (2003) (applied directly, without the inclusion–exclusion step) and exact values to three digits at eight points. Table 2 uses our exact values and adds the two points at which $\hat{s}_1 = 0$. At order one, the approximation of Kolassa and Li is more accurate than ours at seven of the eight points; orders two to five reduce our maximum error to 10^{-3}, 10^{-4}, 10^{-5} and 3×10^{-6}.

At the two points with $\hat{s}_1 < 0$ the values attributed to Kolassa (2003) are his first-order formula evaluated directly at the negative saddlepoint, not his approximation as he prescribes it, which negates that component, differences, and gives our order one (Table 2, footnote). Section 3.3 explains why the direct evaluation has the wrong exponent, and the Supplementary Material follows the two forms as n grows with y fixed: the error of the differenced form

TABLE 1

Wang's example, $n = 10$: relative errors $(\hat{F} - F)/F$ of his simulated values (MC), of his approximation (W) and of the expansions of orders 1 to 5 of $F(x, y) = \mathbb{P}(\bar{X} \le x, \bar{Y} \le y)$, all relative to F itself (notation $a_k = a \times 10^k$). Wang's printed values have one or two significant digits for the smallest probabilities, so the columns MC and W carry rounding errors of up to 2% where two digits are printed and of up to 17% and 22% at $(-1.2, -1.0)$ and $(-1.0, -1.0)$.

(x, y)	exact	MC	W	1	2	3	4	5
$(-1.2, 2.0)$	2.529394_{-4}	2.8_{-2}	6.7_{-2}	-1.4_{-3}	-5.0_{-5}	-4.7_{-7}	2.2_{-8}	8.3_{-10}
$(-1.0, 2.0)$	3.454341_{-3}	1.3_{-2}	4.2_{-2}	-2.0_{-3}	-4.9_{-5}	-1.7_{-7}	2.7_{-8}	4.5_{-10}
$(-0.5, 2.0)$	1.247811_{-1}	1.8_{-3}	3.4_{-2}	-2.7_{-3}	-1.8_{-5}	4.9_{-7}	9.5_{-9}	-5.0_{-10}
$(0.0, 2.0)$	5.297389_{-1}	4.9_{-4}	1.7_{-2}	-1.4_{-5}	1.8_{-6}	1.3_{-8}	-9.2_{-10}	-1.6_{-11}
$(1.5, 2.0)$	9.975398_{-1}	-4.0_{-5}	6.0_{-5}	-7.7_{-7}	9.1_{-8}	2.3_{-9}	1.9_{-11}	-5.2_{-12}
$(2.0, 2.0)$	9.996573_{-1}	-3.7_{-5}	-1.1_{-4}	-5.6_{-7}	6.4_{-9}	3.5_{-10}	1.3_{-11}	-4.1_{-13}
$(0.5, 0.5)$	7.842853_{-1}	-1.6_{-3}	-8.0_{-3}	-1.1_{-4}	6.6_{-6}	3.9_{-7}	-1.3_{-8}	-6.1_{-10}
$(-0.5, 0.0)$	1.038258_{-1}	1.7_{-3}	1.7_{-3}	-4.1_{-3}	-3.5_{-5}	1.5_{-6}	3.1_{-8}	-1.1_{-8}
$(0.0, 0.0)$	3.644369_{-1}	-1.2_{-3}	-1.2_{-2}	-3.8_{-3}	3.9_{-6}	2.0_{-6}	4.6_{-8}	-1.0_{-8}
$(-1.2, -0.5)$	1.725929_{-4}	4.3_{-2}	-1.5_{-2}	-3.6_{-3}	-5.7_{-5}	6.3_{-7}	-1.4_{-7}	-8.6_{-10}
$(-0.5, -0.5)$	4.258299_{-2}	-1.4_{-2}	-1.4_{-2}	-1.4_{-2}	-7.2_{-5}	7.4_{-6}	2.0_{-7}	-3.3_{-8}
$(-0.8, -0.8)$	3.216653_{-3}	2.6_{-2}	-5.2_{-3}	-1.7_{-2}	-2.2_{-4}	9.3_{-6}	5.4_{-7}	-3.5_{-8}
$(-1.2, -1.0)$	2.950726_{-5}	1.7_{-2}	1.7_{-2}	-1.6_{-2}	-2.6_{-4}	1.2_{-5}	6.8_{-7}	-7.6_{-8}
$(-1.0, -1.0)$	2.276964_{-4}	-1.2_{-1}	-1.2_{-1}	-1.7_{-2}	-3.2_{-4}	8.7_{-6}	8.3_{-7}	-2.6_{-8}

TABLE 2

Kolassa and Li's example, $n = 5$: relative errors of their approximation (KL) and of that of Kolassa (2003) (K03), both from their printed values, and of the expansions of orders 1 to 5 of $\mathbb{P}(\bar{Y}_1 \ge y_1, \bar{Y}_2 \ge y_2)$; $\hat{s}$ is the saddlepoint of the cumulant generating function of $(\bar{Y}_1, \bar{Y}_2)$ itself, n times the per-observation saddlepoint of Theorem 1 (notation $a_k = a \times 10^k$). Kolassa and Li omit the last two points, at which $\hat{s}_1 = 0$.

(y_1, y_2)	$\hat{s}$	exact	KL	K03*	1	2	3	4	5
$(2.5, 2.5)$	$(0.65, 0.65)$	9.216039_{-2}	-1.0_{-2}	-2.6_{-2}	-2.6_{-2}	5.2_{-5}	4.0_{-5}	-1.6_{-6}	-4.5_{-8}
$(2.5, 3.5)$	$(-0.52, 2.38)$	1.412745_{-2}	-1.9_{-3}	-1.9_{-3}	-6.5_{-3}	-2.6_{-4}	-1.5_{-5}	-3.7_{-7}	5.0_{-7}
$(2.5, 4.0)$	$(-0.90, 2.87)$	3.933725_{-3}	-6.0_{-3}	1.4_{-2}	-8.3_{-4}	-6.6_{-5}	-2.5_{-5}	-2.8_{-6}	7.4_{-7}
$(3.0, 3.0)$	$(1.06, 1.06)$	2.215196_{-2}	-6.9_{-3}	-3.4_{-2}	-3.3_{-2}	-3.8_{-4}	9.8_{-5}	-1.5_{-6}	-2.1_{-6}
$(3.0, 3.5)$	$(0.48, 1.89)$	8.962830_{-3}	8.0_{-4}	-2.6_{-2}	-2.5_{-2}	-3.7_{-4}	7.7_{-5}	-1.9_{-8}	-2.0_{-6}
$(3.5, 3.5)$	$(1.33, 1.33)$	4.396550_{-3}	7.8_{-4}	-3.3_{-2}	-3.3_{-2}	-8.2_{-4}	1.1_{-4}	8.8_{-6}	-2.8_{-6}
$(3.5, 4.0)$	$(0.80, 2.04)$	1.655068_{-3}	9.0_{-3}	-2.7_{-2}	-2.7_{-2}	-6.3_{-4}	9.5_{-5}	6.6_{-6}	-2.8_{-6}
$(4.0, 4.0)$	$(1.52, 1.52)$	7.583830_{-4}	1.4_{-2}	-3.2_{-2}	-3.2_{-2}	-1.0_{-3}	9.1_{-5}	1.4_{-5}	-1.3_{-6}
$(2.5, 3.0)$	$(0.00, 1.67)$	4.123179_{-2}	—	—	-1.9_{-2}	-2.7_{-4}	4.2_{-5}	4.9_{-7}	-9.7_{-7}
$(3.0, 4.0)$	$(0.00, 2.50)$	2.832675_{-3}	—	—	-1.0_{-2}	-2.4_{-4}	2.8_{-6}	-2.5_{-6}	8.8_{-8}

$*$ At the two points with $\hat{s}_1 < 0$, Kolassa's formula applied without the reduction (17), to match Table 1 of Kolassa and Li (2010); applied as prescribed, it gives our order one at every point.

settles to halving with each doubling of n, while that of the direct form reaches -50% at $n = 640$ and then falls at the rate Δ/n of (22).

The two points that Kolassa and Li omit have $\hat{s}_1 = 0$, where their own approximation has a removable singularity. At these points Theorem 1 applies with a leading term of order $n^{-1/2}$ rather than n^{-1} relative to the exponential factor, and the Supplementary Material shows the relative errors of orders one to five falling at the rates n^{-1} to n^{-5} it predicts at $(2.5, 3.0)$. In double precision their formula loses all accuracy within about 10^{-5} of the lines $\hat{s}_1 = 0$ and $y_2 = E(\bar{Y}_2)$, on which it has the form $\infty - \infty$, and reversing the order of the coordinates, which avoids the singularity, changes its value at the points of Table 2 with $y_1 \ne y_2$ by as much as its error; the Supplementary Material has the details.

5.2. *An example where the approximations of Wang and of Kolassa and Li fail.* The two approximations evaluate the cumulant generating function away from the saddlepoint, at $(0, \hat{s}_2)$ and at $(\hat{s}_1, 0)$, and these points need not lie in its domain; the expansion of Theorem 1 uses K only at $\hat{s}$ and at the saddlepoints of the terms of (17). The example of Section 1 makes the difference visible at ordinary probabilities. Let $X = Z + A$ and $Y = Z + B$ with Z standard Laplace and A, B standard normal, all independent, so that $K(s) = -\log\{1 - (s_1+s_2)^2\} + (s_1^2+s_2^2)/2$ on $|s_1+s_2|<1$. Since $\partial K/\partial s_1 = 2(s_1+s_2)/\{1-(s_1+s_2)^2\} + s_1$, the saddlepoint of a point $(x,-x)$ is $\hat{s} = (x,-x)$ for every n. The approximation of Wang (1990) evaluates K at $(0,\hat{s}_2)$ with the coordinates in the order (X,Y) and at $(\hat{s}_1,0)$ in the order (Y,X), and that of Kolassa and Li (2010, equation (2.6)) at $(\hat{s}_1,0)$ and $(0,\hat{s}_2)$ respectively; both points lie outside the strip when $x \geq 1$, so neither approximation is defined there, in either order and for any n. Our expansion uses K only at $\hat{s}$ and, since $\hat{s}_2 < 0$, at the saddlepoints of the two terms of (17), $\mathbb{P}(\bar{X} \geq x) - \mathbb{P}(\bar{X} \geq x, -\bar{Y} > x)$, the second of which is an orthant probability of $(\bar{X}, -\bar{Y})$ with saddlepoint (x,x), inside the domain $|s_1 - s_2| < 1$ of its cumulant generating function. The exact value is

$$\mathbb{P}(\bar{X} \geq x, \bar{Y} \geq y) = \int_{-\infty}^{\infty} f_{\bar{Z}}(z)\, \bar{\Phi}\{n^{1/2}(x-z)\}\, \bar{\Phi}\{n^{1/2}(y-z)\}\, \mathrm{d}z,$$

with $\bar{\Phi} = 1 - \Phi$ and $n\bar{Z}$ the difference of two independent gamma variables with shape n, whose density is a one-dimensional integral. Table 3 gives the results at five points $(x,-x)$. At the two points where they are defined, $x = 0.4$ and $x = 0.8$, the approximation of Wang (1990) has relative errors of 11% and 23% for $n = 1$ and of 3% and 11% for $n = 10$, and that of Kolassa and Li (2010) of -3% and 25% for $n = 1$ and of 3×10^{-5} and -78% for $n = 10$, against 1%, -3%, -0.4% and -0.8% for the expansion of order one. Two caveats on these values: the term (18) of Wang's Theorem 1 is written as a normal density at $\bar{v}_0$ with $\bar{v}_0^2 < 0$ at every point of the table, and the column W evaluates it through the exponential factor it stands for; and at $x = 0.8$ the correction terms of Kolassa and Li's formula are as large as its leading term, so the formula is defined but no longer meaningful (Supplementary Material). For $n = 1$ the expansion is only asymptotic and gains little beyond order two; for $n = 10$ it behaves as in Section 5.1, and at the point $(1.2, -1.2)$ of Section 1, where the probability is 0.21 for $n = 1$ and 0.015 for $n = 10$, its relative errors fall from 1.1% at order one to 6×10^{-8} at order five.

TABLE 3

Laplace example: $\mathbb{P}(\bar{X} \geq x, \bar{Y} \geq -x)$ and relative errors of the expansions of orders 1 to 5; $\hat{s} = (x,-x)$. W, KL: relative errors of the first-order approximations of Wang (1990), in the order of the coordinates that his rule selects and with the term (18) of his Theorem 1 evaluated through the exponential it stands for (see the text), and of Kolassa and Li (2010), in the order of the coordinates of their paper; — where the approximation evaluates K outside its domain (notation $a_k = a \times 10^k$).

n	$(x,-x)$	exact	W	KL	1	2	3	4	5
1	$(0.4,-0.4)$	3.253461_{-1}	1.1_{-1}	-3.1_{-2}	9.6_{-3}	-4.1_{-3}	8.4_{-4}	2.1_{-3}	-3.7_{-3}
1	$(0.8,-0.8)$	2.753853_{-1}	2.3_{-1}	2.5_{-1}	-3.0_{-2}	-2.3_{-3}	6.5_{-3}	1.5_{-3}	-7.5_{-3}
1	$(1.2,-1.2)$	2.131864_{-1}	—	—	-5.9_{-2}	-5.8_{-3}	8.9_{-3}	4.6_{-3}	-5.4_{-3}
1	$(1.6,-1.6)$	1.548427_{-1}	—	—	-7.7_{-2}	-1.3_{-2}	7.7_{-3}	6.6_{-3}	-4.0_{-4}
1	$(2.0,-2.0)$	1.080995_{-1}	—	—	-8.4_{-2}	-2.0_{-2}	4.5_{-3}	6.4_{-3}	1.8_{-3}
10	$(0.4,-0.4)$	2.243995_{-1}	3.4_{-2}	3.2_{-5}	-4.0_{-3}	7.7_{-5}	1.4_{-6}	-1.5_{-7}	-2.3_{-8}
10	$(0.8,-0.8)$	7.113424_{-2}	1.1_{-1}	-7.8_{-1}	-8.4_{-3}	5.9_{-5}	9.0_{-6}	8.3_{-8}	-9.3_{-8}
10	$(1.2,-1.2)$	1.496196_{-2}	—	—	-1.1_{-2}	-8.2_{-5}	1.1_{-5}	1.1_{-6}	-6.3_{-8}
10	$(1.6,-1.6)$	2.223420_{-3}	—	—	-1.1_{-2}	-2.3_{-4}	2.7_{-6}	1.5_{-6}	6.0_{-8}
10	$(2.0,-2.0)$	2.463991_{-4}	—	—	-9.4_{-3}	-3.0_{-4}	-7.5_{-6}	9.6_{-7}	1.1_{-7}

5.3. *A trivariate example.* Kolassa and Li (2010, Tables 3 and 6) give a conditional and an unconditional example with $d = 3$, both checked against simulation. Here we take their unconditional example and compare with exact values: $Y = (W_{11}, W_{22}, W_{33})$ is the diagonal of a Wishart matrix $W \sim W_3(n, V)$ with $n = 5$ and $V = 0.75 I_3 + 0.25\, 1 1^\top$, at seven points between 0.16 and 0.79 standard deviations above the mean n. The vector Y is the sum of n independent copies of (z_1^2, z_2^2, z_3^2) with $z \sim N(0, V)$, so the setting is that of Theorems 3 and 1 with $\bar{X} = Y/n$, and its cumulant generating function $K(s) = -\frac{n}{2} \log |I_3 - 2V^{1/2} \mathrm{diag}(s) V^{1/2}|$ is that of a vector of three quadratic forms, whose derivatives of all orders follow from the recursion (43) of Section 5.5 below with three matrices. The exact orthant probability and density are one-dimensional integrals. Writing the n columns of the normal matrix as $z_i = 0.5 f_i 1 + 0.75^{1/2} e_i$ with $f_i \sim N(0,1)$ and $e_i \sim N(0, I_3)$ independent, the $Y_j/0.75$ are, given $f = (f_1, \ldots, f_n)$, independent noncentral chi-squared variables with n degrees of freedom and noncentrality $F/3$, where $F = \sum_{i=1}^n f_i^2 \sim \chi^2_n$, so that

$$\mathbb{P}(Y \geq y) = \int_0^\infty p_n(u) \prod_{j=1}^3 \bar{Q}_n(y_j/0.75; u/3)\, \mathrm{d}u,$$

$$f_Y(y) = \int_0^\infty p_n(u) \prod_{j=1}^3 0.75^{-1} q_n(y_j/0.75; u/3)\, \mathrm{d}u,$$

with p_n the χ^2_n density and $\bar{Q}_n(\cdot\,;\lambda)$, $q_n(\cdot\,;\lambda)$ the survival function and the density of the $\chi^2_n(\lambda)$ distribution. The exact values confirm Kolassa and Li's simulation, whose seven estimates lie within 0.8 of their standard errors of the exact values, after two apparent misprints in their table are corrected, the corrections being inferred from the other entries (the third standard error, printed as 1.10×10^{-4}, is 1.10×10^{-3}, and in the last row the normal approximation and the simulation, printed as 1.36×10^{-3} and 1.04×10^{-3}, are 1.36×10^{-2} and 1.04×10^{-2}, as their relative error of -26.9% shows).

Table 4 gives the relative errors. The approximation of Kolassa and Li has relative errors of -13% to -26%, and the approximation of Kolassa (2003), our order one, has -16% to -27%: with $n = 5$ the first-order error is large whichever construction is used. Order two reduces the error to 0.2–1.4%, order three to at most 0.16%, and orders four and five to 10^{-6}–2×10^{-4}; the density behaves in the same way. With $n = 20$ and the points at the same standardized offsets, $y = n + (y^{(5)} - 5)(n/5)^{1/2}$, the errors fall from 3–6% at order one to 10^{-9}–6×10^{-8} at order five, the pattern of the bivariate examples. The integrals $I(\nu)$, $|\nu| \leq 27$ at order five, were computed as described in Appendix C; the cancellation indicator $\mathcal{C}$ of the orthant probability expansion is at most 1.8×10^4 for $n = 5$ and 36 for $n = 20$, and the quadrature check of Appendix C, with 128 nodes per coordinate, at most 3×10^{-12} relative, far below the printed errors, so the third dimension does not reduce the accuracy here.

5.4. *A ratio of regression coefficients whose denominator takes both signs.* Bebu, Seillier-Moiseiwitsch and Mathew (2009) construct generalized confidence intervals for a ratio β_i/β_j of coefficients of a normal linear regression from the quantiles of $R = (\hat{\beta}_i - \hat{\sigma}_i \tilde{T}_1)/(\hat{\beta}_j - \hat{\sigma}_j \tilde{T}_2)$, where $\hat{\beta}$ and $\hat{\sigma}$ are the estimates and their standard errors and $(\tilde{T}_1, \tilde{T}_2)$ is bivariate Student t with ν degrees of freedom, the residual degrees of freedom of the regression, and the estimated correlation ρ of $(\hat{\beta}_i, \hat{\beta}_j)$. As Broda and Kan (2016) show, $R = \tilde{R} + q$ with $q = \rho \hat{\sigma}_i/\hat{\sigma}_j$ and $\tilde{R} = (a + bT_1)/(c + dT_2)$, $T_k = Z_k (X/\nu)^{-1/2}$, where Z_1, Z_2 are standard normal and $X \sim \chi^2_\nu$, all independent, and $a = \hat{\beta}_i - q\hat{\beta}_j$, $b = -\hat{\sigma}_i (1 - \rho^2)^{1/2}$, $c = \hat{\beta}_j$,

TABLE 4

Trivariate example, the diagonal of a $W_3(n,V)$ matrix: relative errors of the approximation of Kolassa and Li (2010) (KL, from their printed values, for $n=5$ only) and of the expansions of orders 1 to 5 of the orthant probability $\mathbb{P}(Y \geq y)$ and of the density, at the points of Kolassa and Li's Table 6 for $n=5$ and at the points with the same standardized offsets for $n=20$ (notation $a_k = a \times 10^k$).

y	exact	KL	1	2	3	4	5
$n=5$, orthant probability							
$(5.5,5.5,5.5)$	5.650358_{-2}	-1.3_{-1}	-1.6_{-1}	-1.9_{-3}	4.0_{-4}	6.1_{-5}	1.1_{-5}
$(5.5,5.5,6.5)$	4.258540_{-2}	-1.4_{-1}	-1.8_{-1}	-2.9_{-3}	5.6_{-4}	4.9_{-5}	9.2_{-6}
$(5.5,6.5,6.5)$	3.223587_{-2}	-1.6_{-1}	-2.0_{-1}	-4.2_{-3}	7.5_{-4}	3.2_{-5}	1.3_{-6}
$(6.5,6.5,6.5)$	2.450839_{-2}	-1.9_{-1}	-2.3_{-1}	-5.8_{-3}	1.0_{-3}	1.4_{-5}	-1.6_{-5}
$(6.5,6.5,7.5)$	1.827417_{-2}	-2.1_{-1}	-2.4_{-1}	-8.2_{-3}	1.2_{-3}	4.4_{-5}	-4.8_{-5}
$(6.5,7.5,7.5)$	1.368481_{-2}	-2.3_{-1}	-2.5_{-1}	-1.1_{-2}	1.4_{-3}	9.7_{-5}	-8.9_{-5}
$(7.5,7.5,7.5)$	1.029240_{-2}	-2.6_{-1}	-2.7_{-1}	-1.4_{-2}	1.6_{-3}	1.8_{-4}	-1.4_{-4}
$n=5$, density							
$(5.5,5.5,5.5)$	1.332410_{-3}		1.1_{-1}	-6.4_{-3}	-3.7_{-4}	1.5_{-4}	1.2_{-5}
$(5.5,5.5,6.5)$	1.045248_{-3}		1.1_{-1}	-6.4_{-3}	-4.0_{-4}	1.6_{-4}	2.3_{-5}
$(5.5,6.5,6.5)$	8.249296_{-4}		1.1_{-1}	-6.4_{-3}	-4.5_{-4}	1.7_{-4}	3.6_{-5}
$(6.5,6.5,6.5)$	6.551154_{-4}		1.1_{-1}	-6.4_{-3}	-5.0_{-4}	1.7_{-4}	5.2_{-5}
$(6.5,6.5,7.5)$	5.013887_{-4}		1.1_{-1}	-6.4_{-3}	-5.5_{-4}	1.6_{-4}	6.7_{-5}
$(6.5,7.5,7.5)$	3.859986_{-4}		1.1_{-1}	-6.4_{-3}	-6.0_{-4}	1.5_{-4}	8.1_{-5}
$(7.5,7.5,7.5)$	2.989625_{-4}		1.1_{-1}	-6.3_{-3}	-6.5_{-4}	1.3_{-4}	9.2_{-5}
$n=20$, orthant probability							
$(21,21,21)$	7.451290_{-2}		-3.1_{-2}	-6.4_{-5}	5.1_{-6}	1.3_{-7}	9.6_{-9}
$(21,21,23)$	5.574538_{-2}		-3.5_{-2}	-1.2_{-4}	7.1_{-6}	1.2_{-7}	9.2_{-9}
$(21,23,23)$	4.185346_{-2}		-4.0_{-2}	-1.9_{-4}	9.5_{-6}	8.8_{-8}	6.8_{-9}
$(23,23,23)$	3.153522_{-2}		-4.6_{-2}	-2.6_{-4}	1.2_{-5}	4.5_{-8}	1.6_{-9}
$(23,23,25)$	2.281637_{-2}		-4.9_{-2}	-3.9_{-4}	1.4_{-5}	1.1_{-7}	-1.3_{-8}
$(23,25,25)$	1.657151_{-2}		-5.1_{-2}	-5.2_{-4}	1.7_{-5}	2.0_{-7}	-3.4_{-8}
$(25,25,25)$	1.208204_{-2}		-5.5_{-2}	-6.6_{-4}	1.9_{-5}	3.3_{-7}	-6.1_{-8}
$n=20$, density							
$(21,21,21)$	2.060802_{-4}		2.6_{-2}	-3.5_{-4}	-6.5_{-6}	5.0_{-7}	2.0_{-8}
$(21,21,23)$	1.729413_{-4}		2.6_{-2}	-3.5_{-4}	-6.8_{-6}	5.2_{-7}	2.6_{-8}
$(21,23,23)$	1.459899_{-4}		2.6_{-2}	-3.5_{-4}	-7.1_{-6}	5.3_{-7}	3.3_{-8}
$(23,23,23)$	1.239799_{-4}		2.6_{-2}	-3.6_{-4}	-7.5_{-6}	5.4_{-7}	4.0_{-8}
$(23,23,25)$	9.824587_{-5}		2.6_{-2}	-3.6_{-4}	-7.8_{-6}	5.4_{-7}	4.8_{-8}
$(23,25,25)$	7.830824_{-5}		2.6_{-2}	-3.6_{-4}	-8.2_{-6}	5.3_{-7}	5.6_{-8}
$(25,25,25)$	6.278675_{-5}		2.6_{-2}	-3.5_{-4}	-8.6_{-6}	5.2_{-7}	6.4_{-8}

$d=-\hat{\sigma}_j$. The Student variables have no cumulant generating function, but multiplying numerator and denominator by X/ν gives $\tilde{R}=\tilde{X}/\tilde{Y}$ with $\tilde{X}=aX/\nu+bZ_1(X/\nu)^{1/2}$ and $\tilde{Y}=cX/\nu+dZ_2(X/\nu)^{1/2}$, whose joint cumulant generating function is

$$K(s,t)=-\frac{\nu}{2}\log\left\{1-\frac{b^2s^2+d^2t^2+2(as+ct)}{\nu}\right\}; \tag{42}$$

writing $X=\sum_{k=1}^{\nu}N_k^2$ and $Z(X/\nu)^{1/2}=\nu^{-1/2}\sum_k N_kM_k$ with independent standard normal N_k, M_k shows that $(\tilde{X},\tilde{Y})$ is a pair of quadratic forms in a Gaussian vector of dimension 3ν with ν identical 3×3 blocks. The denominator $\tilde{Y}$ takes both signs, and by Broda and Kan (2016, Lemma 2), $\mathbb{P}(\tilde{R}\leq r)=\mathbb{P}(W<0)+\mathbb{P}(Y<0)-2\mathbb{P}(W<0,Y<0)$ with $W=\tilde{X}-r\tilde{Y}$ and $Y=\tilde{Y}$, again quadratic forms. The first two terms are given by the expansions of Section 4 and the third by Theorem 1 for $(-W,-Y)$ at 0, through (17) when its saddlepoint has a negative component. The pair $(\tilde{X},\tilde{Y})$ is the mean of the ν independent pairs $(aN_k^2+b\nu^{1/2}N_kM_k,\, cN_k^2+d\nu^{1/2}N_kL_k)$, L_k being a further set of independent standard normal variables, whose cumulant generating function $-\frac{1}{2}\log\{1-2(as+ct)-\nu(b^2s^2+d^2t^2)\}$

satisfies (1) with $n_0 = 3$; for fixed a and c and standard errors $b = b_0\nu^{-1/2}$ and $d = d_0\nu^{-1/2}$ with fixed $b_0, d_0 \neq 0$, the scaling of the usual asymptotics of a regression, this distribution does not depend on ν and Theorems 1 and 2 apply with $n = \nu$; a general sequence of regressions gives a triangular array, which the theorems cover only through uniform versions of their conditions, not pursued here. The relative error bounds for the three terms of $\mathbb{P}(\tilde{R} \leq r)$ carry over to their combination only up to the factor of Section 3.3, the sum of the terms over the result, which is moderate in the example below but is not claimed to be bounded in general. Given X the pair is bivariate normal, so $\mathbb{P}(W < 0)$ and $\mathbb{P}(Y < 0)$ are Student t probabilities with ν degrees of freedom and $\mathbb{P}(W < 0, Y < 0)$ is a bivariate t probability; the exact values are these, computed as bivariate normal orthant probabilities given X integrated against the χ^2_ν density.

Bebu, Seillier-Moiseiwitsch and Mathew (2009) and Broda and Kan (2016) analyze Cox's (1985) data on the percentage of fine gravel in two kinds of surface soil, for which $\hat{\beta}_1 = 10.91$, $\hat{\beta}_2 = 3.94$, $\hat{\sigma}_1 = \hat{\sigma}_2 = 1.83$, $\rho = 0$ and $\nu = 12$ (the programs fit the regression to the data and use the unrounded estimates), so that $\mathbb{P}(Y < 0) = 2.6\%$ and Fieller's interval is not a proper interval at the 95% level. Table 5 shows the relative errors of the expansions of orders one to five with (17). The order-one approximation is that of Broda and Kan (2016), whose Theorem 5 handles a saddlepoint with mixed signs by the reflection and the marginal term of (17). For the 95% interval Broda and Kan (2016) report $(-3.310, 15.680)$ exactly and $(-2.240, 15.370)$ from their approximation; orders two, three and four give $(-3.297, 15.678)$, $(-3.312, 15.680)$ and $(-3.310, 15.680)$, the fourth agreeing with the exact interval to three decimals.

TABLE 5

Ratio of regression coefficients with a denominator of both signs, Cox's gravel data ($\nu = 12$, $\mathbb{P}(Y < 0) = 2.6\%$): exact $\mathbb{P}(\tilde{R} \leq r)$, and the relative errors, normalized by $\min(F, 1 - F)$, of the expansions of orders 1 to 5 with (17), the first of which is the approximation of Broda and Kan (2016) (notation $a_k = a \times 10^k$). The saddlepoint has a negative second component for $r < -0.36$.

r	exact	1	2	3	4	5
-5	2.299627_{-2}	-3.3_{-2}	-5.2_{-4}	7.9_{-5}	3.8_{-6}	-9.3_{-7}
-3	2.529930_{-2}	-3.4_{-2}	-5.7_{-4}	7.9_{-5}	4.3_{-6}	-9.4_{-7}
-1	2.626994_{-2}	-3.4_{-2}	-5.9_{-4}	8.0_{-5}	4.6_{-6}	-9.4_{-7}
0	2.633404_{-2}	-3.4_{-2}	-6.0_{-4}	8.0_{-5}	4.6_{-6}	-9.4_{-7}
2	2.634757_{-1}	-1.7_{-2}	2.7_{-6}	2.9_{-5}	1.5_{-7}	-2.8_{-7}
5	8.437208_{-1}	1.6_{-2}	-1.3_{-4}	-2.6_{-5}	1.0_{-6}	2.1_{-7}
15	9.734595_{-1}	2.6_{-2}	1.4_{-4}	-5.9_{-5}	-4.0_{-7}	6.2_{-7}

5.5. *Pairs of quadratic forms in normal vectors.* Let $z \sim N(\mu, I_N)$ and let $Q_1 = z^\top Az$, $Q_2 = z^\top Bz$ be two quadratic forms. The joint cumulant generating function is $K(s_1, s_2) = -\frac{1}{2}\log|I - 2C| + \frac{1}{2}\mu^\top\{(I - 2C)^{-1} - I\}\mu$ with $C = s_1A + s_2B$, and its derivatives of any order at $s = (s_1, s_2)^\top$ are, with $D = (I - 2C)^{-1}$,

$$\frac{\partial^{|\nu|}K(s)}{\partial s_1^{\nu_1}\partial s_2^{\nu_2}} = \frac{2^{|\nu|-1}\nu!}{|\nu|}\left\{\operatorname{tr}(F_\nu) + |\nu|\,\mu^\top F_\nu D\mu\right\}, \tag{43}$$

where $F_\nu = DA\,F_{\nu-e_1} + DB\,F_{\nu-e_2}$, $F_0 = I$, is the sum of all products of ν_1 copies of DA and ν_2 copies of DB in every order. The bivariate density and distribution function expansions of Section 3 then apply with $d = 2$, and the orthant probabilities needed are the bivariate normal integral and univariate normal tail probabilities. The joint distribution of two

quadratic forms arises, for example, as that of two sums of squares in a linear model; its exact evaluation requires a two-dimensional numerical inversion.

To obtain exact values to many digits we use a class of pairs with a common component. Let $A = \mathrm{diag}(I_{m_0}, I_{m_1}, 0_{m_2})$ and $B = \mathrm{diag}(I_{m_0}, 0_{m_1}, I_{m_2})$, so that $Q_1 = X_0 + X_1$ and $Q_2 = X_0 + X_2$ with $X_i \sim \chi^2_{m_i}(\lambda_i)$, $i = 0, 1, 2$, independent, X_0 being the common component. With $\lambda_0 = \lambda_1 = \lambda_2 = 0$ and $m_0 = m_1 = m_2$ even, this class contains, up to location and scale, the bivariate gamma distribution used as the example by Wang (1990) and Kolassa and Li (2010). Then $K(s_1, s_2) = K_1(s_1) + K_2(s_2) + K_0(s_1 + s_2)$ with $K_i(s) = -\frac{m_i}{2}\log(1 - 2s) + \lambda_i s/(1 - 2s)$ the cumulant generating function of $\chi^2_{m_i}(\lambda_i)$, and the density and the orthant probability of (Q_1, Q_2) are one-dimensional integrals over X_0 of products of noncentral chi-squared densities and tail probabilities, which we evaluated to 25 digits. The correlation of Q_1 and Q_2 is about 0.5 in the two cases of Table 6. The six points are at $E(Q_i) + c_i\,\mathrm{sd}(Q_i)$ for the (c_1, c_2) shown. At the points marked with an asterisk the saddlepoint has a negative first component although both coordinates are above their means, and the orthant probability was obtained from (17) as $\mathbb{P}(Q_2 \geq x_2) - \mathbb{P}(-Q_1 \geq -x_1, Q_2 \geq x_2)$, the univariate term from Corollary 2. The expansions were evaluated in the whitened coordinates of Appendix C, where the cancellation indicator $\mathcal{C}$ is at most 4, and agree with a computation in 40-digit arithmetic to 7×10^{-14} relative; in the original coordinates $\mathcal{C}$ reaches 2×10^6, which would leave an accuracy comparable to the smallest errors of order five.

TABLE 6

Pairs of quadratic forms with a common component: relative errors of the expansions of order M of the orthant probability $\mathbb{P}(Q_1 \geq x_1, Q_2 \geq x_2)$ and of the density at $x_i = E(Q_i) + c_i\,\mathrm{sd}(Q_i)$ (notation $a_k = a \times 10^k$). Asterisk: saddlepoint with a negative component, orthant probability by (17).

	orthant probability, order M					density, order M				
(c_1, c_2)	1	2	3	4	5	1	2	3	4	5
$(m_0, m_1, m_2) = (10, 15, 20)$, $(\lambda_0, \lambda_1, \lambda_2) = (8, 5, 10)$										
(0.5,0.5)	-9.4_{-3}	-2.3_{-5}	4.7_{-6}	2.6_{-7}	-2.4_{-8}	-8.9_{-4}	-6.7_{-4}	-1.1_{-5}	6.9_{-6}	1.4_{-6}
(1.0,1.5)	-1.1_{-2}	-8.1_{-5}	8.1_{-6}	2.5_{-7}	-1.2_{-7}	8.9_{-4}	-2.7_{-4}	3.7_{-5}	7.9_{-6}	9.5_{-8}
(1.5,1.0)	-1.2_{-2}	-1.7_{-4}	5.1_{-6}	8.1_{-7}	1.7_{-8}	-3.1_{-3}	-9.7_{-4}	-6.7_{-5}	-1.4_{-6}	7.6_{-7}
(2.0,2.0)	-1.3_{-2}	-2.3_{-4}	8.0_{-6}	1.3_{-6}	-1.5_{-8}	-2.5_{-3}	-6.5_{-4}	-6.0_{-6}	7.4_{-6}	1.4_{-6}
(1.0,3.0)*	-6.3_{-4}	1.8_{-5}	-2.7_{-7}	-5.1_{-7}	-2.2_{-8}	8.6_{-3}	3.0_{-4}	1.8_{-5}	-3.3_{-6}	-3.3_{-7}
(3.0,3.0)	-1.1_{-2}	-2.5_{-4}	6.2_{-6}	1.4_{-6}	-7.3_{-9}	-3.4_{-3}	-6.2_{-4}	7.6_{-7}	7.9_{-6}	1.3_{-6}
$(m_0, m_1, m_2) = (30, 40, 60)$, $(\lambda_0, \lambda_1, \lambda_2) = (30, 20, 10)$										
(0.5,0.5)	-3.2_{-3}	-3.8_{-6}	1.7_{-7}	4.6_{-9}	5.5_{-11}	-1.7_{-3}	-1.4_{-4}	-3.7_{-6}	-8.6_{-8}	-2.0_{-10}
(1.0,1.5)	-4.0_{-3}	-1.7_{-5}	2.6_{-7}	1.3_{-8}	1.8_{-10}	-1.9_{-3}	-1.3_{-4}	-3.0_{-6}	-1.1_{-8}	5.6_{-9}
(1.5,1.0)	-4.0_{-3}	-2.0_{-5}	1.3_{-7}	8.5_{-9}	1.4_{-10}	-1.7_{-3}	-1.3_{-4}	-3.3_{-6}	-8.6_{-8}	-1.8_{-9}
(2.0,2.0)	-5.1_{-3}	-3.5_{-5}	1.6_{-7}	1.9_{-8}	4.8_{-10}	-2.2_{-3}	-1.4_{-4}	-3.9_{-6}	-9.7_{-8}	-9.6_{-10}
(1.0,3.0)*	-1.8_{-4}	-6.8_{-6}	8.8_{-9}	4.1_{-9}	-1.4_{-10}	-4.9_{-4}	-3.9_{-5}	1.8_{-6}	1.4_{-7}	1.0_{-9}
(3.0,3.0)	-4.7_{-3}	-4.5_{-5}	-2.1_{-8}	2.1_{-8}	8.3_{-10}	-2.5_{-3}	-1.4_{-4}	-4.0_{-6}	-1.0_{-7}	-1.3_{-9}

For the larger case the relative error of the orthant probability expansion improves from about 10^{-3} at order one to about 10^{-10} at order five, and that of the density from about 10^{-3} to 10^{-9}; for the smaller case the orthant probability reaches 10^{-7}–10^{-8} at order five and the density, whose gains diminish after order three, about 10^{-6}, as for the univariate distributions with small degrees of freedom. The relative errors at the mixed-sign points are no worse than at the others, so the inclusion–exclusion step does not affect the accuracy.

6. Conclusion. The saddlepoint expansion of the multivariate orthant probability has been given to arbitrary order. Its relative error is uniform over compact subsets of the closed positive orthant of saddlepoints, including those with vanishing components. Two recursions

generate the terms of every order, one on the Taylor coefficients of the cumulant generating function at the saddlepoint, which also generates the density expansion and the expansions for ratios, and one for the derivatives of the normal orthant probability. Saddlepoints outside the positive orthant are handled by an inclusion–exclusion identity, given in closed form, which preserves the relative error bound under conditions on the saddlepoints of the marginals, and we have shown that this step cannot in general be dispensed with. The expansion has no singular terms and does not depend on the order of the coordinates, properties it shares with the first-order approximation of Kolassa (2003), its truncation at order one. As demonstrated in the applications, the gain from including higher-order terms is substantial unless the sample size or the degrees of freedom are very small.

Two extensions are natural. The first is to variables on a unit lattice, handled as in Kolassa (2003) and Kolassa (2006, Sections 5.8 and 6.9): the inversion is taken over one period in each lattice coordinate, s_j in the denominator of (9) is replaced by $2\sinh(s_j/2)$, and the threshold of an inclusive tail $\mathbb{P}(S \geq k)$ of a unit-lattice sum is moved to $k - 1/2$, since $e^{-ks}/(1-e^{-s}) = e^{-(k-1/2)s}/\{2\sinh(s/2)\}$; the factor $s_j/\{2\sinh(s_j/2)\}$ then enters the integrand and is expanded about the saddlepoint together with the cumulant generating function. The second is to conditional distributions. Skovgaard (1987) provides a Lugannani–Rice type approximation for the conditional distribution, with the second-order term given in Kolassa (1996) (see also Kolassa and Li, 2010). Our approach would yield an expansion of Hauschildt–Robinson type, in which the numerator leads to integrals (11) with some components of r equal to 0 instead of -1, already covered by Lemma 2, and the denominator is a marginal density expansion; only the components of the saddlepoint that correspond to tail probabilities carry a pole at the origin, so the sign restriction and the reduction (17) apply to those components alone.

APPENDIX A: RATIOS

A.1. Vectors of ratios. Let (X_i, Y_i), $i = 1, \ldots, n$, be independent and identically distributed pairs of d-vectors with joint cumulant generating function $\mathbb{K}(s,t)$, $s,t \in \mathbb{R}^d$, with $Y > 0$ almost surely componentwise, and let $R = (\bar{X}_1/\bar{Y}_1, \ldots, \bar{X}_d/\bar{Y}_d)$. Ratios with a common denominator are the case $Y_1 = \cdots = Y_d$, in which $\mathbb{K}$ depends on t only through $t_1 + \cdots + t_d$. With $W_j = X_j - r_j Y_j$ and $r \circ s = (r_1 s_1, \ldots, r_d s_d)$, the cumulant generating function of W is $K_W(s) = \mathbb{K}(s, -r \circ s)$, and $\mathbb{P}(R \leq r) = \mathbb{P}(\bar{W} \leq 0) = \mathbb{P}(-\bar{W} \geq 0)$ is the orthant probability of $-\bar{W}$ at 0, whose cumulant generating function is $K_W(-s)$. The joint distribution function of R is therefore given by Theorem 1 with $\bar{x} = 0$ and saddlepoint $-\hat{s}$, where $K'_W(\hat{s}) = 0$, and by (17) when $-\hat{s}$ has negative components. A component $\hat{s}_j = 0$ is handled as in Section 3.3.

The joint density is obtained by differentiating

$$\mathbb{P}(\bar{W} \leq 0) = \frac{1}{(2\pi\mathrm{i})^d} \int_{s_0 - \mathrm{i}\infty}^{s_0 + \mathrm{i}\infty} \exp\{nK_W(s)\} \prod_{j=1}^{d} \frac{\mathrm{d}s_j}{-s_j}, \qquad s_0 < 0, \tag{44}$$

with respect to $r_1, \ldots, r_d$. For $B \subseteq \{1, \ldots, d\}$ let

$$\mathbb{K}_B = \frac{\partial^{|B|}\mathbb{K}}{\prod_{j \in B} \partial t_j}.$$

Since $\partial K_W / \partial r_j = -s_j \mathbb{K}_{\{j\}}(s, -r \circ s)$ and the derivative of $\mathbb{K}_B(s, -r \circ s)$ with respect to r_j is $-s_j \mathbb{K}_{B \cup \{j\}}$, each differentiation cancels one pole and the usual formula for the derivative of $\exp\{nK_W\}$ with respect to d distinct variables gives

$$f_R(r) = \frac{1}{(2\pi\mathrm{i})^d} \int_{\hat{s} - \mathrm{i}\infty}^{\hat{s} + \mathrm{i}\infty} e^{nK_W(s)} g_n(s)\, \mathrm{d}s, \qquad g_n(s) = \sum_{\pi \in \mathcal{P}_d} n^{|\pi|} \prod_{B \in \pi} \mathbb{K}_B(s, -r \circ s), \tag{45}$$

where $\mathcal{P}_d$ is the set of partitions π of $\{1,\dots,d\}$ and $|\pi|$ is the number of blocks, so that $g_n = \sum_{k=0}^{d-1} n^{d-k} g_k$ with g_k collecting the partitions into $d-k$ blocks; the poles having disappeared, the contour passes through the saddlepoint $\hat{s}$ of K_W, which need not lie in any particular orthant. For $d=2$, $g_n = n^2 \mathbb{K}_{\{1\}}\mathbb{K}_{\{2\}} + n\mathbb{K}_{\{1,2\}}$. For $n=1$, (45) is the inversion of the moment generating function in Butler (2007, Section 12.3), and $g_0(\hat{s}) = \prod_{j=1}^{d} \mathbb{K}_{\{j\}}(\hat{s}, -r\circ\hat{s})$ is the factor in the first-order density of vector ratios of Butler and Paolella (1998) and Butler (2007, equations (12.49)–(12.51)).

Equation (45) is a saddlepoint integral with a non-constant amplitude, and it is expanded by combining (4) with the Taylor expansion of the amplitude.

THEOREM 4. *Let ψ be a cumulant generating function satisfying (1), $\hat{s}$ its saddlepoint, $\psi'(\hat{s})=0$, $c_{r,\nu}$ the coefficients (4) of its remainder G at $\hat{s}$, with $c_{0,0}=1$, and let h be analytic in a neighborhood of $\operatorname{Re} s = \hat{s}$, with $h(\hat{s}+t) = \sum_{l=0}^{\infty}\sum_{|\mu|=l}\gamma_\mu t^\mu$ near $\hat{s}$ and*

$$\int_{\mathbb{R}^d} |h(\hat{s}+\mathrm{i}u)|\,|M_{\hat{s}}(u)|^{n_0}\,\mathrm{d}u < \infty, \tag{46}$$

$M_{\hat{s}}(u) = \exp\{\psi(\hat{s}+\mathrm{i}u) - \psi(\hat{s})\}$, uniformly for $\hat{s}$ in compact sets. Then

$$\frac{1}{(2\pi\mathrm{i})^d}\int_{\hat{s}-\mathrm{i}\infty}^{\hat{s}+\mathrm{i}\infty} e^{n\psi(s)}h(s)\,\mathrm{d}s = \frac{|\psi''(\hat{s})|^{-1/2}e^{n\psi(\hat{s})}}{(2\pi n)^{d/2}}\left\{\sum_{m=0}^{M-1}\frac{b_m}{n^m} + O(n^{-M})\right\},$$

$$b_m = \sum_{r=0}^{2m}\sum_{j=3r}^{2m+2r}\sum_{|\nu|=j}\sum_{|\mu|=2m+2r-j} c_{r,\nu}\gamma_\mu \mathrm{He}_{\nu+\mu}\{0,\psi''(\hat{s})\}. \tag{47}$$

The proof is that of Theorem 3 with the additional factor $h(\hat{s}+\mathrm{i}y/\sqrt{n})$: the term with (r,ν,μ) carries $n^{r-(|\nu|+|\mu|)/2}$ and the moment $\mathrm{i}^{|\nu+\mu|}E(S^{\nu+\mu})$, which vanishes unless $|\nu|+|\mu|$ is even, the Taylor remainder of h is bounded as in Lemma 3, and (46) bounds the part of the contour with $|y|>\delta\sqrt{n}$ by ρ^{n-n_0} times a constant. Condition (46) cannot be replaced by a polynomial bound on h: integrability of $|M_{\hat{s}}|^{n_0}$ does not imply that of $|u|^2|M_{\hat{s}}(u)|^n$ for any n. The term with $m=1$ is the correction of Kolassa (2006, Lemma 6.5.3) for real Laplace integrals with an amplitude. For $h\equiv 1$ only $\mu=0$ survives and (47) is (24). Applying (47) to (45) with $\psi = K_W$ and $h=g_k$, writing $b_j(g_k)$ for its coefficients, gives the following.

COROLLARY 3. *Let K_W and $h=g_k$, $k=0,\dots,d-1$, satisfy the conditions of Theorem 4. Then*

$$f_R(r) = \left(\frac{n}{2\pi}\right)^{\frac{d}{2}}|\hat{K}_W''|^{-\frac{1}{2}}e^{nK_W(\hat{s})}\left\{\sum_{m=0}^{M-1}\frac{\tilde{a}_m}{n^m} + O(n^{-M})\right\},$$

$$\tilde{a}_m = \sum_{k=0}^{\min(m,d-1)} b_{m-k}(g_k), \tag{48}$$

The leading coefficient is $\tilde{a}_0 = g_0(\hat{s})$, and the term with k is the contribution of the partitions with $d-k$ blocks, which enter k orders later than the product of first derivatives; the derivatives of $\mathbb{K}$ required are available in closed form in the examples below, where the g_k are bounded on the contour and (46) holds. For $d\geq 2$, the approximation with $M=1$, $\tilde{a}_0 = g_0(\hat{s})$, is that of Butler and Paolella (1998, Section 3) for the serial correlogram, who retain only the leading term of the saddlepoint expansion and, in the amplitude, only the product of first derivatives, the other terms being of relative order n^{-1} (their Section 10); Butler

(2007, equations (12.49)–(12.51)) gives it for general vector ratios. The terms with $M \geq 2$, which bring in the mixed derivatives of $\mathbb{K}$ through $g_1, \ldots, g_{d-1}$ as well as the higher-order terms of the saddlepoint expansion, are new.

A.2. One ratio. For $d = 1$, $R = \bar{X}/\bar{Y}$ and $K_W(s) = \mathbb{K}(s, -rs)$. Let $\hat{s}$ solve $K'_W(\hat{s}) = 0$, write $\hat{K}_{W,k} = K_W^{(k)}(\hat{s})$, and let $\hat{w}$, $\hat{u}$, the θ_k, $d_{r,m}$ and $a_{i,j}$ be those of (26)–(29) with K_W in place of K and $\bar{x} = 0$. Since $\mathbb{P}(R \leq r) = \mathbb{P}(\bar{W} \leq 0)$, Corollary 2 with $\bar{x} = 0$ gives the distribution function.

COROLLARY 4. *Let $p \geq 0$ be an integer. For $\hat{s} < 0$, $\mathbb{P}(R < r)$ is the right-hand side of (37), and for $\hat{s} > 0$, $\mathbb{P}(R > r)$ is the right-hand side of (36).*

For the density, (45) with $d = 1$ is (Geary, 1944; Broda and Kan, 2016)

$$f_R(r) = \frac{n}{2\pi \mathrm{i}} \int_{\hat{s}-\mathrm{i}\infty}^{\hat{s}+\mathrm{i}\infty} e^{nK_W(s)}\, g(s)\, \mathrm{d}s, \qquad g(s) = \mathbb{K}_2(s, -rs) = \frac{\partial \mathbb{K}(s,t)}{\partial t}\Big|_{t=-rs}, \tag{49}$$

with $g(\hat{s}) = E(Ye^{\hat{s}W})/E(e^{\hat{s}W}) > 0$. With the standardized coefficients

$$\tilde{\gamma}_l = \frac{g^{(l)}(\hat{s})}{l!\, g(\hat{s}) \hat{K}_{W,2}^{l/2}}, \qquad \tilde{a}_j = \sum_{k=0}^{2j} \tilde{\gamma}_{2j-k}\, a_{j,k}, \tag{50}$$

where the $a_{j,k}$ are computed from the θ_k of K_W at $\hat{s}$ and $\tilde{\gamma}_0 = \tilde{a}_0 = 1$, Corollary 3 with $d = 1$, in which only $k = 0$ remains, $g_0 = g$ and $\gamma_\mu = \tilde{\gamma}_\mu g(\hat{s}) \hat{K}_{W,2}^{\mu/2}$, gives the following, with its coefficients divided by $g(\hat{s})$.

COROLLARY 5 (Rice). *If K_W and g satisfy the conditions of Theorem 4, then for $M \geq 1$,*

$$f_R(r) = \frac{\sqrt{n}\, g(\hat{s})\, \phi(\sqrt{n}\hat{w})}{\sqrt{\hat{K}_{W,2}}} \left\{ \sum_{j=0}^{M-1} \frac{\tilde{a}_j}{n^j} + O(n^{-M}) \right\}, \qquad \hat{w} = \mathrm{sgn}(\hat{s}) \sqrt{-2K_W(\hat{s})}. \tag{51}$$

The leading term is the approximation of Daniels (1954, Section 9). The second-order coefficient is

$$\tilde{a}_1 = a_1 + 3\theta_3\tilde{\gamma}_1 - \tilde{\gamma}_2 = 3\theta_4 - \tfrac{15}{2}\theta_3^2 + 3\theta_3\tilde{\gamma}_1 - \tilde{\gamma}_2$$

as given in Butler and Paolella (2007, equations 20 and 21) and Butler (2007, p. 383). For $g \equiv 1$ this is Corollary 1. The coefficient $\tilde{a}_j$ is a convolution of the density coefficients $a_{j,k}$ with the standardized derivatives of the amplitude: the term $g^{(2j-k)}$ pairs with $a_{j,k}$ because $g^{(l)}(\hat{s})(s - \hat{s})^l/l!$ raises the degree by l without adding a factor of n. The expansion (51) is that of Rice (1968, equation (103)) with its terms regrouped by powers of n^{-1} and written with the θ_k in place of the derivatives of the exponent. Alternatively, the truncated expansion (36) applied to $\bar{W}$ at $\bar{x} = 0$ can be differentiated with respect to r, since $\hat{s}$, $\hat{w}$ and the θ_k are smooth functions of r; this is a formal operation, as the derivative of an $O(n^{-M})$ remainder need not be $O(n^{-M})$, and the error of the resulting density approximation is established through (51), from which it differs by terms of relative order n^{-M}; it is regular at $\hat{s} = 0$, the point $r = E(X)/E(Y)$, whereas the derivative of (41) inherits the singularity of (41) there. Equation (51) itself involves only $\phi(\sqrt{n}\hat{w})$ and polynomials in the θ_k and $\tilde{\gamma}_l$, and needs only the derivatives of K_W and g at the saddlepoint. The Supplementary Material applies Corollaries 4 and 5 to the singly and doubly noncentral F distributions, $F = (Q_1/m)/(Q_2/n)$ with $Q_1 \sim \chi^2_m(\lambda_1)$ and $Q_2 \sim \chi^2_n(\lambda_2)$ independent, at 1.01×10^7 and 1.6×10^7 points: the median relative error at order six is 8×10^{-11}–4×10^{-13} and 4×10^{-10}–2×10^{-13}, and with a denominator with $n \leq 2$ the maximum error stops improving after order three or four.

APPENDIX B: PROOFS

The proofs of Theorems 1 and 3 rest on a bound for the remainder of the expansion (4) after the terms with $m = |\nu| - 2r \le p$. Let T be a compact subset of $\mathcal{T}$, let $\lambda > 0$ be the smallest eigenvalue of $\hat{K}''$ over $\hat{s} \in T$, and let $\bar{\kappa}_k = \sup_{\hat{s}\in T} \sum_{|\nu|=k} |\kappa_\nu|$; p is a nonnegative integer.

LEMMA 3. *There are $\delta > 0$ and $C < \infty$, depending only on K, T and p, such that for all $\hat{s} \in T$, all n and all real or complex y with $|y| \le \delta\sqrt{n}$,*

$$\Big|\exp\{nG(\mathrm{i}y/\sqrt{n})\} - \sum_{m=0}^{p} n^{-m/2} \sum_{r=0}^{m} \sum_{|\nu|=m+2r} c_{r,\nu}(\mathrm{i}y)^\nu\Big| \le C n^{-(p+1)/2}|y|^{p+1} e^{\lambda|y|^2/4}, \tag{52}$$

where the term with $m = r = 0$ is 1.

PROOF. Since K is analytic on a complex neighborhood of T, there is $\delta_0 > 0$ such that K is analytic and bounded on the polydisc of radius $3\delta_0$ about every $\hat{s} \in T$, and Cauchy's estimates give $|\kappa_\nu| \le M(3\delta_0)^{-|\nu|}$ with M independent of $\hat{s} \in T$; as there are at most $(k+1)^{d-1}$ multi-indices ν with $|\nu| = k$, $\sum_{k=3}^{\infty} \bar{\kappa}_k (2\delta_0)^k < \infty$, so the Taylor series of K at $\hat{s}$ converges absolutely on the polydisc of radius $2\delta_0$, uniformly for $\hat{s} \in T$, and

$$\eta(\delta) = \sum_{k=3}^{\infty} \bar{\kappa}_k \delta^{k-2}$$

decreases to 0 as $\delta \to 0$. Choose $\delta \le \delta_0$ with $\eta(\delta) \le \lambda/4$. By (4), $\sum_{|\nu|=j} |c_{r,\nu}|$ is at most the coefficient of z^j in $(\sum_{k=3}^{\infty} \bar{\kappa}_k z^k)^r / r!$. For $|y| \le \delta\sqrt{n}$ the double series

$$\sum_{r=0}^{\infty} \sum_{k=3r}^{\infty} \sum_{|\nu|=k} n^{r-|\nu|/2} c_{r,\nu}(\mathrm{i}y)^\nu$$

converges absolutely to $\exp\{nG(\mathrm{i}y/\sqrt{n})\}$, so the left side of (52) is at most

$$\sum_{r=0}^{\infty} \sum_{m=p+1}^{\infty} \sum_{|\nu|=m+2r} n^{-m/2} |c_{r,\nu}| |y|^{|\nu|}.$$

In a term of this sum

$$n^{-m/2}|y|^{|\nu|} = n^{-(p+1)/2}|y|^{2r+p+1}\Big(\frac{|y|}{\sqrt{n}}\Big)^{m-p-1} \le n^{-(p+1)/2}|y|^{2r+p+1}\delta^{|\nu|-2r-p-1},$$

and summing over ν and then over r,

$$\sum_{r=0}^{\infty} n^{-(p+1)/2}\Big(\frac{|y|}{\delta}\Big)^{p+1}\Big(\frac{|y|^2}{\delta^2}\Big)^r \frac{\{\sum_{k=3}^{\infty} \bar{\kappa}_k \delta^k\}^r}{r!} = n^{-(p+1)/2}\Big(\frac{|y|}{\delta}\Big)^{p+1} \exp\{\eta(\delta)|y|^2\},$$

which gives (52) with $C = \delta^{-p-1}$. The argument uses only the absolute convergence of the series and $|(\mathrm{i}y)^\nu| \le |y|^{|\nu|}$, so it applies to complex y with $|y|$ the Euclidean norm. □

PROOF OF LEMMA 1. Write B for the complement of A, $u = (u_A, u_B)$, and E_s for the expectation under the distribution tilted by s. The domain of K_A is the intersection of $\mathcal{T}$ with the coordinate subspace $\{s : s_B = 0\}$, open in that subspace and containing the origin, and K''_A is a principal submatrix of K'', hence positive definite. The characteristic function of the tilted marginal is $M_s(u_A, 0)$ with $s = (s_A, 0)$, so for a compact set T_A in the domain of

K_A both conditions in (1) for K_A concern M_s on the compact set $T = T_A \times \{0\} \subset \mathcal{T}$. Since $|(u_A, 0)| = |u_A|$, the second condition for K_A is the second condition for K on a subset. For the first, $|e^{\mathrm{i}\theta} - 1| \le |\theta|$ gives, for $s \in T$,

$$|M_s(u_A, u_B) - M_s(u_A, 0)| = \left|E_s\{e^{\mathrm{i}u_A^\top X_A}(e^{\mathrm{i}u_B^\top X_B} - 1)\}\right| \le |u_B|\, E_s|X_B| \le C|u_B|,$$

where $C = \sup_{s\in T} E_s|X_B|$ is finite because $E_s X_j^2 = K_{jj}''(s) + K_j'(s)^2$ is continuous on $\mathcal{T}$. Hence, with $g(u_A) = |M_s(u_A, 0)|$, $|M_s(u_A, u_B)| \ge g(u_A)/2$ whenever $|u_B| \le g(u_A)/(2C)$, and integrating over this ball,

$$\int_{\mathbb{R}^d} |M_s(u)|^{n_0}\,\mathrm{d}u \ge V_{|B|} \int_{\mathbb{R}^{|A|}} \left\{\frac{g(u_A)}{2}\right\}^{n_0} \left\{\frac{g(u_A)}{2C}\right\}^{|B|} \mathrm{d}u_A,$$

with V_k the volume of the unit ball of $\mathbb{R}^k$. Thus $\sup_{s\in T} \int_{\mathbb{R}^{|A|}} |M_s(u_A, 0)|^{n_0+|B|}\,\mathrm{d}u_A$ is at most a constant times $\sup_{s\in T} \int_{\mathbb{R}^d} |M_s(u)|^{n_0}\,\mathrm{d}u$, which is finite. □

Let $\tilde{f}_n$ be the density of the mean of n observations from the distribution tilted by $\hat{s}$, whose mean is $\bar{x}$ and whose covariance matrix is $\hat{K}''$ for one observation and $\hat{K}''/n$ for the mean, let $\bar{f}_n(v) = n^{-d/2}\tilde{f}_n(\bar{x} + v/\sqrt{n})$, the density of $\sqrt{n}(\bar{X} - \bar{x})$ under the tilted distribution, with covariance matrix $\hat{K}''$, and let

$$P_m(v) = \sum_{r=0}^{m} \sum_{|\nu|=m+2r} c_{r,\nu}(-D)^\nu \phi(v, \hat{K}''), \qquad D^\nu = \frac{\partial^{|\nu|}}{\partial v_1^{\nu_1} \cdots \partial v_d^{\nu_d}},$$

so that $P_0 = \phi(\cdot, \hat{K}'')$ and P_m, the term of order $n^{-m/2}$ of the Edgeworth expansion of $\bar{f}_n$, is a polynomial of degree $3m$ times $\phi(\cdot, \hat{K}'')$.

LEMMA 4. *There are n_1 and C, depending only on K, T and p, such that for all $n \ge n_1$, all $\hat{s} \in T$ and all $v \in \mathbb{R}^d$,*

$$\prod_{j=1}^{d}(1 + v_j^2)\left|\bar{f}_n(v) - \sum_{m=0}^{p} n^{-m/2} P_m(v)\right| \le C n^{-(p+1)/2}.$$

PROOF. Let $\tilde{M}_{\hat{s}}(u) = \exp\{K(\hat{s} + \mathrm{i}u) - \hat{K} - \mathrm{i}u^\top \bar{x}\}$ be the characteristic function of the tilted distribution centered at its mean, so that $|\tilde{M}_{\hat{s}}| = |M_{\hat{s}}|$ and, by (3),

$$\tilde{M}_{\hat{s}}(y/\sqrt{n})^n = \exp\{-y^\top \hat{K}'' y/2 + nG(\mathrm{i}y/\sqrt{n})\}.$$

For $n \ge n_0$, Fourier inversion gives

$$\bar{f}_n(v) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-\mathrm{i}v^\top y} \tilde{M}_{\hat{s}}(y/\sqrt{n})^n\,\mathrm{d}y,$$

$$\frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-\mathrm{i}v^\top y} (\mathrm{i}y)^\nu e^{-y^\top \hat{K}'' y/2}\,\mathrm{d}y = (-D)^\nu \phi(v, \hat{K}''),$$

so the difference in the lemma is $(2\pi)^{-d} \int_{\mathbb{R}^d} e^{-\mathrm{i}v^\top y} F_n(y)\,\mathrm{d}y$ with

$$F_n(y) = \tilde{M}_{\hat{s}}(y/\sqrt{n})^n - e^{-y^\top \hat{K}'' y/2} S_p(y),$$

where $S_p(y)$ is the sum in (52); by the display above, for $|y| \le 2\delta_0\sqrt{n}$, where the Taylor series of G converges,

$$F_n(y) = e^{-y^\top \hat{K}'' y/2}\big[\exp\{nG(\mathrm{i}y/\sqrt{n})\} - S_p(y)\big],$$

the form used on the inner region below. Let $L = \prod_{j=1}^d (1 - \partial^2/\partial y_j^2)$. Since F_n and its derivatives of order up to $2d$ are integrable, as shown below, integration by parts gives $\prod_{j=1}^d (1+v_j^2) \int_{\mathbb{R}^d} e^{-\mathrm{i}v^\top y} F_n(y)\,\mathrm{d}y = \int_{\mathbb{R}^d} e^{-\mathrm{i}v^\top y} L F_n(y)\,\mathrm{d}y$, and it suffices to show that $\int_{\mathbb{R}^d} |LF_n(y)|\,\mathrm{d}y \le Cn^{-(p+1)/2}$ uniformly on T. Let δ be as in Lemma 3 and let Λ bound the largest eigenvalue of $\hat{K}''$ on T.

On $|y| \le \delta\sqrt{n}/2$ we use Lemma 3 with complex arguments. Let a be real with $|a| \le \delta\sqrt{n}/2$ and $y = a + w$ with $|w_j| \le 1$ for all j, so that $|y| \le |a| + \sqrt{d} \le \delta\sqrt{n}$ once $n \ge 4d/\delta^2$, and write $a' = a + \operatorname{Re} w$ and $b = \operatorname{Im} w$, so that $|a' - a| \le \sqrt{d}$ and $|b| \le \sqrt{d}$. Then $\operatorname{Re}(y^\top \hat{K}'' y) = {a'}^\top \hat{K}'' a' - b^\top \hat{K}'' b \ge \lambda |a'|^2 - \Lambda d$, and (52) gives

$$\begin{aligned} |F_n(y)| &\le Cn^{-(p+1)/2}(|a'| + \sqrt{d})^{p+1} \exp\{\lambda(|a'| + \sqrt{d})^2/4 - \lambda|a'|^2/2 + \Lambda d/2\} \\ &\le C' n^{-(p+1)/2}(1 + |a'|)^{p+1} e^{-\lambda|a'|^2/8} \le C'' n^{-(p+1)/2}(1+|a|)^{p+1} e^{-\lambda|a|^2/16}, \end{aligned}$$

the last step because $|a'| \le |a| + \sqrt{d}$ and $|a'|^2 \ge (|a| - \sqrt{d})_+^2 \ge |a|^2/2 - d$. The points of the polydisc $\{y : |y_j - a_j| \le 1,\ j = 1, \dots, d\}$ are of this form, so by Cauchy's estimates every derivative of F_n of order at most $2d$ obeys the last bound at a with a larger constant, so $\int_{|a| \le \delta\sqrt{n}/2} |LF_n(a)|\,\mathrm{d}a \le Cn^{-(p+1)/2}$.

On $|y| > \delta\sqrt{n}/2$, the derivatives of the terms of F_n that are polynomials times $e^{-y^\top \hat{K}'' y/2}$ integrate to $O(e^{-\lambda\delta^2 n/16})$, and a derivative of order $k \le 2d$ of $\tilde{M}_{\hat{s}}(y/\sqrt{n})^n$ with respect to y is $n^{-k/2}$ times a derivative of order k of $\tilde{M}_{\hat{s}}^n$ at $u = y/\sqrt{n}$, which is a sum of at most $C_k n^k$ terms, each $\tilde{M}_{\hat{s}}^{n-j}$ with $j \le k$ times a product of j derivatives of $\tilde{M}_{\hat{s}}$ of total order k. The derivatives of $\tilde{M}_{\hat{s}}$ are bounded by absolute moments of the tilted distribution, which are bounded uniformly on T, and $|\tilde{M}_{\hat{s}}| \le 1$, so for $|\alpha| \le 2d$

$$|D^\alpha \{\tilde{M}_{\hat{s}}(y/\sqrt{n})^n\}| \le Cn^{|\alpha|/2} |M_{\hat{s}}(y/\sqrt{n})|^{n-|\alpha|}.$$

With $\rho = \sup\{|M_{\hat{s}}(u)| : |u| \ge \delta/2, \hat{s} \in T\} < 1$ by (1), for $n \ge n_0 + 2d$,

$$\begin{aligned} \int_{|y| > \delta\sqrt{n}/2} |D^\alpha \{\tilde{M}_{\hat{s}}(y/\sqrt{n})^n\}|\,\mathrm{d}y &\le Cn^{|\alpha|/2} \int_{|y| > \delta\sqrt{n}/2} |M_{\hat{s}}(y/\sqrt{n})|^{n-|\alpha|}\,\mathrm{d}y \\ &\le Cn^{(|\alpha|+d)/2} \rho^{n-|\alpha|-n_0} \sup_{\hat{s}\in T} \int_{\mathbb{R}^d} |M_{\hat{s}}(u)|^{n_0}\,\mathrm{d}u, \end{aligned}$$

which is exponentially small; the same bound over $\mathbb{R}^d$ shows that F_n and its derivatives of order up to $2d$ are integrable. □

PROOF OF THEOREM 1. Let T be a compact subset of $\mathcal{T}$ in the closed positive orthant, λ and Λ the bounds on the eigenvalues of $\hat{K}''$ over T, and $\hat{u} = \sqrt{n}\hat{s}$. The density of $\bar{X}$ is $e^{n(\hat{K} - \hat{s}^\top x)} \tilde{f}_n(x)$, so for $n \ge n_0$, with $x = \bar{x} + v/\sqrt{n}$,

$$\mathbb{P}(\bar{X} \ge \bar{x}) = e^{n(\hat{K} - \hat{s}^\top \bar{x})} \int_{[0,\infty)^d} e^{-\hat{u}^\top v} \bar{f}_n(v)\,\mathrm{d}v,$$

the tilting representation of Esscher (1932), as used by Robinson (1982) and Daniels (1987, Section 3), valid for every $\hat{s}$ in the closed orthant. Two elementary bounds on Laplace transforms over the orthant are needed: for $\hat{u} \ge 0$, with C depending only on d, λ and Λ,

$$\int_{[0,\infty)^d} e^{-\hat{u}^\top v} \prod_{j=1}^d \frac{\mathrm{d}v_j}{1+v_j^2} \le C \prod_{j=1}^d \frac{1}{1+\hat{u}_j},$$

$$C^{-1} \prod_{j=1}^d \frac{1}{1+\hat{u}_j} \le \int_{[0,\infty)^d} e^{-\hat{u}^\top v} \phi(v, \hat{K}'')\,\mathrm{d}v \le C \prod_{j=1}^d \frac{1}{1+\hat{u}_j}.$$

The first integral factorizes, and $\int_0^\infty e^{-uv}(1+v^2)^{-1}\mathrm{d}v \le \min(\pi/2, 1/u) \le (1+\pi/2)/(1+u)$. For the second, $\Lambda^{-1}|v|^2 \le v^\top(\hat{K}'')^{-1}v \le \lambda^{-1}|v|^2$ brackets the integral between constant multiples of $\prod_{j=1}^d \int_0^\infty e^{-\hat{u}_j v - v^2/(2\lambda)}\mathrm{d}v$ and $\prod_{j=1}^d \int_0^\infty e^{-\hat{u}_j v - v^2/(2\Lambda)}\mathrm{d}v$, and for $c > 0$, $\int_0^\infty e^{-uv-cv^2}\mathrm{d}v \le \min\{(\pi/4c)^{1/2}, 1/u\}$ while $\int_0^\infty e^{-uv-cv^2}\mathrm{d}v \ge e^{-1}\int_0^{c^{-1/2}} e^{-uv}\mathrm{d}v \ge e^{-1}(1-e^{-1})\min(c^{-1/2}, 1/u)$, both between constant multiples of $(1+u)^{-1}$.

By Lemma 4, for $n \ge n_1$,

$$\bar{f}_n = \sum_{m=0}^{p} n^{-m/2} P_m + R_n, \qquad |R_n(v)| \le C n^{-(p+1)/2} \prod_{j=1}^d \frac{1}{1+v_j^2},$$

so in the representation the remainder contributes at most $Cn^{-(p+1)/2} e^{n(\hat{K}-\hat{s}^\top\bar{x})} \prod_{j=1}^d (1+\hat{u}_j)^{-1}$, the term $m=0$ contributes $e^{n(\hat{K}-\hat{s}^\top\bar{x})}\int_{[0,\infty)^d} e^{-\hat{u}^\top v}\phi(v,\hat{K}'')\,\mathrm{d}v$, of exact order $e^{n(\hat{K}-\hat{s}^\top\bar{x})}\prod_{j=1}^d(1+\hat{u}_j)^{-1}$, and, since $(-D)^\nu\phi(v,\hat{K}'')$ is a polynomial times $\phi(v,\hat{K}'')$ and hence bounded by $C_\nu\prod_{j=1}^d(1+v_j^2)^{-1}$, the term with (r,ν) contributes at most

$$C_\nu n^{-m/2} e^{n(\hat{K}-\hat{s}^\top\bar{x})} \prod_{j=1}^d \frac{1}{1+\hat{u}_j}.$$

It remains to identify the terms with those of (12). For $\hat{s}$ in the open orthant, $1/(\hat{u}_j + \mathrm{i}y_j) = \int_0^\infty e^{-(\hat{u}_j+\mathrm{i}y_j)v_j}\mathrm{d}v_j$ and Fubini's theorem give

$$\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d} e^{-y^\top\hat{K}''y/2}(\mathrm{i}y)^\nu \prod_{j=1}^d \frac{\mathrm{d}y_j}{\hat{u}_j+\mathrm{i}y_j} = \int_{[0,\infty)^d} e^{-\hat{u}^\top v}(-D)^\nu\phi(v,\hat{K}'')\,\mathrm{d}v,$$

and the substitution $s = \hat{s} + \mathrm{i}y/\sqrt{n}$ in (11), for which $(s-\hat{s})^\nu = n^{-|\nu|/2}(\mathrm{i}y)^\nu$, $\prod_{j=1}^d \mathrm{d}s_j/s_j = \prod_{j=1}^d \mathrm{i}\,\mathrm{d}y_j/(\hat{u}_j+\mathrm{i}y_j)$ and the exponent equals $-\hat{u}^\top\hat{K}''\hat{u}/2 - y^\top\hat{K}''y/2$, shows that the left side is $n^{|\nu|/2}e^{nq/2}I(\nu)$ with $q = \hat{s}^\top\hat{K}''\hat{s}$. Since $E_n = e^{n(\hat{K}-\hat{s}^\top\bar{x})}e^{nq/2}$ and $n^{r-|\nu|/2} = n^{-m/2}$, the term with (r,ν) is $E_n n^r c_{r,\nu} I(\nu)$ and the term $m=0$ is $E_n I(0)$. The identity $n^{|\nu|/2}e^{nq/2}I(\nu) = \int_{[0,\infty)^d} e^{-\hat{u}^\top v}(-D)^\nu\phi(v,\hat{K}'')\,\mathrm{d}v$ has so far been proved for $\hat{s}$ in the open orthant, where the contour in (11) avoids the poles; both sides are continuous in $\hat{s}$ on the closed orthant, the left through (14) and (15) and the right by dominated convergence, so it extends to the boundary with $I(\nu)$ given by (14). The terms with $m \ge 1$ are $O(n^{-1/2})$ times the leading term by the bounds above, so the bracket in (12) is $I(0)\{1+O(n^{-1/2})\}$, and dividing by E_n times the bracket gives (12) with the error uniform on T. Finally, the identity and the bounds on the Laplace transforms give the bounds on $I(\nu)$ and $I(0)$ stated after the theorem. □

PROOF OF LEMMA 2. Write $u = t_\mathcal{I} - \Sigma_{\mathcal{I}\mathcal{J}}\Sigma_{\mathcal{J}\mathcal{J}}^{-1}t_\mathcal{J}$ and $g = \phi(t_\mathcal{J}, \Sigma_{\mathcal{J}\mathcal{J}})\bar{\Phi}(u, \Sigma_{\mathcal{I}\mathcal{I}\cdot\mathcal{J}})$ as in (15). For $i \in \mathcal{I}$, D_i acts only on $\bar{\Phi}$, with $\partial u_i/\partial t_i = 1$, so

$$J(r+e_i) = -\prod_{j\in\mathcal{J}}(-D_j)^{r_j}\,\phi\,D_i\bar{\Phi}.$$

For $j \in \mathcal{J}$, $-D_j\phi(t_\mathcal{J}, \Sigma_{\mathcal{J}\mathcal{J}}) = (\Sigma_{\mathcal{J}\mathcal{J}}^{-1}t_\mathcal{J})_{(j)}\phi$ and, since $\partial u_i/\partial t_j = -(\Sigma_{\mathcal{I}\mathcal{J}}\Sigma_{\mathcal{J}\mathcal{J}}^{-1})_{(i,j)}$,

$$D_j\bar{\Phi} = -\sum_{i\in\mathcal{I}}(\Sigma_{\mathcal{I}\mathcal{J}}\Sigma_{\mathcal{J}\mathcal{J}}^{-1})_{(i,j)}D_i\bar{\Phi}.$$

Hence

$$-D_j g = (\Sigma_{\mathcal{J}\mathcal{J}}^{-1} t_{\mathcal{J}})_{(j)} g + \phi \sum_{i\in\mathcal{I}} (\Sigma_{\mathcal{I}\mathcal{J}}\Sigma_{\mathcal{J}\mathcal{J}}^{-1})_{(i,j)} D_i \bar{\Phi},$$

and applying $\prod_{k\in\mathcal{J}}(-D_k)^{r_k}$,

$$J(r+e_j) = \prod_{k\in\mathcal{J}} (-D_k)^{r_k} \big[(\Sigma_{\mathcal{J}\mathcal{J}}^{-1} t_{\mathcal{J}})_{(j)} g\big] - \sum_{i\in\mathcal{I}} (\Sigma_{\mathcal{I}\mathcal{J}}\Sigma_{\mathcal{J}\mathcal{J}}^{-1})_{(i,j)} J(r+e_i).$$

The factor $(\Sigma_{\mathcal{J}\mathcal{J}}^{-1} t_{\mathcal{J}})_{(j)}$ is linear in $t_{\mathcal{J}}$ with $-D_k(\Sigma_{\mathcal{J}\mathcal{J}}^{-1} t_{\mathcal{J}})_{(j)} = -(\Sigma_{\mathcal{J}\mathcal{J}}^{-1})_{(j,k)}$, so the Leibniz rule for a linear factor, $(-D_k)^{r_k}(fg) = f(-D_k)^{r_k} g + r_k(-D_k f)(-D_k)^{r_k-1} g$, applied for each $k \in \mathcal{J}$ with $r_k > 0$, gives the first two terms of (16). □

PROOF OF THEOREM 2. Write $P = \mathbb{P}(\bar{X} \geq \bar{x})$, $s^0 = s^{(\emptyset)}$ and, for $j \in N$, $R_j = \mathbb{P}(\bar{X}_{N^c} \geq \bar{x}_{N^c}, \bar{X}_j \leq \bar{x}_j)$. Since $P_T = \mathbb{P}(\bar{X}_{N^c} \geq \bar{x}_{N^c}, \bar{X}_T \leq \bar{x}_T)$ and $\bar{X}$ has a density for $n \geq n_0$, so that the events $\{\bar{X}_j = \bar{x}_j\}$ have probability zero, $P_T \leq P_\emptyset$ for every T, $P_T \leq R_j$ for every nonempty T and $j \in T$, and $P_\emptyset - P = \mathbb{P}(\bar{X}_{N^c} \geq \bar{x}_{N^c},\ \bar{X}_j < \bar{x}_j \text{ for some } j \in N) \leq \sum_{j\in N} R_j$; the event of P_T is contained in that of $P_\emptyset - P$ for the same reason, so $P_T \leq P_\emptyset - P$. On a compact set of $\bar{x}$ on which (a) holds, the saddlepoints $s^{(T)}$ depend continuously on $\bar{x}$, being the inverses of the diffeomorphisms $K'_{N^c\cup T}$ on $\mathcal{S}_T \cap \mathcal{T}$, and so range over compact subsets of $\mathcal{T}$; Theorem 1 applied to each $X^{(T)}$ under (a) and Lemma 1 is therefore uniform on the set. Let $\gamma_j = K'_j(s^0) - \bar{x}_j$ for $j \in N$, which is nonnegative under (a), as shown at the end of the proof, and let $\bar{x}^0 = K'(s^0)$, so that $\bar{x}^0_{N^c} = \bar{x}_{N^c}$ and $\bar{x}^0_N = \bar{x}_N + \gamma_N \geq \bar{x}_N$. Then $P \geq P^0 = \mathbb{P}(\bar{X} \geq \bar{x}^0)$, and the saddlepoint of $\bar{x}^0$ is s^0, which lies in the closed positive orthant by (a) for $T = \emptyset$ and has $s^0_N = 0$. Theorem 1 with $p = 0$ at $\bar{x}^0$ gives $P^0 = E_n I(0)\{1 + O(n^{-1/2})\}$ with $E_n = \exp\{n(K(s^0) - s^{0\top}\bar{x}^0 + s^{0\top}K''(s^0)s^0/2)\}$ and $I(0) = \mathbb{P}(Z \geq \sqrt{n}K''(s^0)s^0)$, $Z \sim N(0, K''(s^0))$; applied to $X^{(\emptyset)}$ when $N^c \neq \emptyset$ it gives $P_\emptyset = E_n I^{(\emptyset)}(0)\{1 + O(n^{-1/2})\}$ with the same factor E_n, since $s^0_N = 0$, and $I^{(\emptyset)}(0) = \mathbb{P}(Z_{N^c} \geq \sqrt{n}(K''(s^0)s^0)_{N^c})$, while $P_\emptyset = 1 = E_n = I^{(\emptyset)}(0)$ and $s^0 = 0$ when $N^c = \emptyset$. By the bounds following Theorem 1, $I(0)$ and $I^{(\emptyset)}(0)$ are both of exact order $e^{-nq/2}\prod_{j\in N^c}(1 + \sqrt{n}s^0_j)^{-1}$ with $q = s^{0\top}K''(s^0)s^0$, the factors for $j \in N$ being 1 because $s^0_N = 0$, so $I(0)/I^{(\emptyset)}(0) \geq \delta > 0$ with δ depending only on the compact set. Hence $P \geq P^0 \geq \delta P_\emptyset/2$ for n large, so $\sum_{T\subseteq N} P_T \leq 2^{|N|} P_\emptyset \leq 2^{|N|+1}\delta^{-1}P$, the first assertion, and since $|P_T - \hat{P}_T| \leq C_M n^{-M} P_T$ uniformly by Theorem 1, (17) and (19) give

$$|P - \hat{P}(\bar{x})| \leq \sum_{T\subseteq N} |P_T - \hat{P}_T| \leq C_M n^{-M} \sum_{T\subseteq N} P_T \leq 2^{|N|+1}\delta^{-1} C_M n^{-M} P,$$

which is (20). Now let (b) hold, so that $\gamma_j > 0$ for $j \in N$. Let $\ell_0 = K(s^0) - s^{0\top}\bar{x}$, which is 0 when $N^c = \emptyset$. For $\varepsilon > 0$ with $s^0 - \varepsilon e_j \in \mathcal{T}$, the exponent $n\{\sum_{i\in N^c} s^0_i(\bar{X}_i - \bar{x}_i) - \varepsilon(\bar{X}_j - \bar{x}_j)\}$ is nonnegative on the event of R_j, since $s^0_{N^c} \geq 0$ by (a), so by Markov's inequality

$$\begin{aligned} R_j &\leq E\exp\{n(s^0 - \varepsilon e_j)^\top(\bar{X} - \bar{x})\} = \exp[n\{K(s^0 - \varepsilon e_j) - (s^0 - \varepsilon e_j)^\top \bar{x}\}] \\ &= \exp\{n(\ell_0 - \varepsilon\gamma_j + \tfrac{1}{2}\varepsilon^2 K''_{jj}(\xi))\} \end{aligned}$$

for some ξ on the segment from $s^0 - \varepsilon e_j$ to s^0. On a compact set of $\bar{x}$ on which (a) and (b) hold, γ_j is bounded away from zero, by the continuity of s^0, and K''_{jj} is bounded on a neighborhood of the s^0, so ε can be chosen with $\varepsilon K''_{jj}(\xi) \leq \gamma_j$, giving $R_j \leq e^{n(\ell_0 - 2c)}$ with $2c = \inf_{\bar{x}} \min_{j\in N} \varepsilon\gamma_j/2 > 0$. If $N^c = \emptyset$ then $P_\emptyset = 1 = e^{n\ell_0}$; otherwise Theorem 1 with $p = 0$, applied to $X^{(\emptyset)}$ under (a) and Lemma 1, gives $P_\emptyset = E_n I(0)\{1 + O(n^{-1/2})\}$ with $E_n I(0)$ of

exact order $e^{n\ell_0}\prod_{j\in N^c}(1+\sqrt{n}s_j^0)^{-1}\geq C^{-1}e^{n\ell_0}n^{-|N^c|/2}$, so that $P_\emptyset\geq C^{-1}n^{-|N^c|/2}e^{n\ell_0}$ for n large. In either case $R_j\leq Cn^{|N^c|/2}e^{-2cn}P_\emptyset\leq e^{-cn}P_\emptyset/|N|$ for n large, which proves the exponential comparison; the ratio $\sum_{T\subseteq N}P_T/P$ then lies between 1 and $(1+2^{|N|}e^{-cn})/(1-e^{-cn})$. The argument for the exponential comparison applies verbatim with the minimizer s^* of $K(s)-s^\top\bar{x}$ over the closed positive orthant in place of s^0 and $A=\{j:s_j^*>0\}$ in place of N^c, since it used only $s_A^*>0$, $K'_A(s^*)=\bar{x}_A$ and $K'_j(s^*)>\bar{x}_j$ for $j\notin A$; this gives the statement following the theorem. Finally, let $j\in N$, $f(s)=K(s)-s^\top\bar{x}$, which is strictly convex on the interior of the convex set on which K is finite, $u=s^{(\{j\})}$ and $v=s^{(\emptyset)}$. Both lie in $\mathcal{S}_{\{j\}}$, $v_j=0$, the partial derivatives of f vanish at u for the coordinates in $N^c\cup\{j\}$ and at v for those in N^c, and $f'_j(v)=K'_j(s^{(\emptyset)})-\bar{x}_j=\gamma_j$. The gradient of a strictly convex function is strictly monotone, so if $u\neq v$ then $0<\{f'(v)-f'(u)\}^\top(v-u)=-u_j\gamma_j$, only the j-th term of the sum being nonzero. By (a) for $T=\{j\}$, $u_j\leq 0$. If $u_j<0$ then $u\neq v$ and $\gamma_j>0$; if $u_j=0$ then u solves the saddlepoint equations of X_{N^c}, so $u=v$ and $\gamma_j=0$. Hence $\gamma_j\geq 0$ under (a) alone, and $\gamma_j>0$ if and only if $s_j^{(\{j\})}<0$, which is the equivalence stated after the theorem. When $N=\{j\}$ and $\hat{s}_j<0$, $s^{(\{j\})}=\hat{s}$, which gives (b). □

Continuity of (19) across a change of N. Consider first a vector with saddlepoint $\hat{s}$, $\hat{s}_j=0$, $\hat{s}_i\geq 0$ for $i\neq j$, and the three expansions of Theorem 1 of the same order: $\hat{P}$ for its orthant probability, $\hat{P}'$ for that of the marginal without coordinate j, and $\hat{P}''$ for that of the vector with coordinate j reflected. The saddlepoints of the marginal and of the reflected vector are $\hat{s}$ itself, the former because $\hat{s}$ solves the equations of the coordinates $i\neq j$ with $s_j=0$ and the latter because reflecting the j-th component of $\hat{s}$, which is zero, leaves $\hat{s}$ unchanged; so the three share the factor E_n. The coefficients $c_{r,\nu}$ of the marginal are those of K with $\nu_j=0$, since setting $t_j=0$ in the remainder G of K at $\hat{s}$ gives the remainder of the marginal's cumulant generating function, and those of the reflected vector are $(-1)^{\nu_j}c_{r,\nu}$. In the orthant-integral form of the proof of Theorem 1, with $\hat{u}_j=0$, the integral of the reflected term over its orthant is, after $v_j\to -v_j$, that of $(-1)^{\nu_j}(-D)^\nu\phi(v,\hat{K}'')e^{-\hat{u}^\top v}$ over $\{v_j\leq 0\}$, so with its coefficient it adds to the term of $\hat{P}$ to give the integral over the whole v_j-axis, which is 0 for $\nu_j\geq 1$ and the marginal's integral for $\nu_j=0$. Hence $\hat{P}'-\hat{P}''=\hat{P}$ term by term, at every order. Now let $\bar{x}$ cross a boundary $\hat{s}_j=0$ at which N becomes $N'=N\cup\{j\}$, with (a) holding on both sides. For $T\subseteq N$ the terms T and $T\cup\{j\}$ of (19) for N' solve the saddlepoint equations of the term T for N with coordinate j dropped and reflected respectively; (a) for N requires $s_j^{(T)}\geq 0$ before the crossing and (a) for N' requires the same quantity to be ≤ 0 after it, so by continuity $s_j^{(T)}=0$ at the crossing, and the identity just proved, applied to the vector of the term T, gives $\hat{P}_T^{(N')}-\hat{P}_{T\cup\{j\}}^{(N')}=\hat{P}_T^{(N)}$. Summing over $T\subseteq N$ with the signs $(-1)^{|T|}$ gives $\hat{P}^{(N')}=\hat{P}^{(N)}$ at the crossing, and the continuity of the $I(\nu)$ in $\hat{s}$ on the closed orthant makes the two approximations join continuously. When some $s_j^{(T)}$ does not vanish at the crossing, (a) fails on one side and the formula changes there.

Continuity of the recursive application of (17). Let $F(Q)$ denote the approximation that the recursion produces for an orthant probability Q at a fixed order: the expansion of Theorem 1 when the saddlepoint of Q has no negative component, and otherwise $\sum_{S\subseteq N_Q}(-1)^{|S|}F(Q_S)$, where N_Q is the set of negative components and the Q_S are the terms of (17) for Q with the index set N_Q; $F=1$ for the empty vector, and $F(Q)$ is defined where all the saddlepoints that the recursion requires exist. For a coordinate j of Q let Q' and Q'' be the marginal without j and the reflection of j, as above. We show that $F(Q)=F(Q')-F(Q'')$ for every Q and j. Adding an index k to an index set N replaces each term Q_S of (17) by $Q'_S-Q''_S$ with respect to k, so the identity implies

that $\sum_{S\subseteq N}(-1)^{|S|}F(Q_S) = F(Q)$ for every index set N, not only for N_Q; this is what makes the recursion independent of the choices made at a boundary. The proof is by induction on the dimension d of Q, the consequence just drawn being available in dimension $d-1$. Let u be the saddlepoint of Q. If $u_j > 0$, the negative set of Q'' is $N_Q \cup \{j\}$, and in (17) for Q'' the terms S and $S\cup\{j\}$, $S \subseteq N_Q$, are Q'_S and Q_S itself, so $F(Q'') = \sum_{S\subseteq N_Q}(-1)^{|S|}\{F(Q'_S) - F(Q_S)\} = \sum_{S\subseteq N_Q}(-1)^{|S|}F(Q'_S) - F(Q)$, where the first sum is $F(Q')$ by the consequence in dimension $d-1$, the Q'_S being the terms of (17) for Q' with the index set N_Q; when $N_Q = \emptyset$ this is the definition of $F(Q'')$. If $u_j < 0$, the case just treated applies to Q'', whose j-th saddlepoint component is $-u_j > 0$ and whose marginal and reflection are Q' and Q. If $u_j = 0$, then Q'' has the saddlepoint u and the negative set N_Q, so $F(Q)$, $F(Q'')$ and, by the consequence in dimension $d-1$, $F(Q')$ are the alternating sums over $S \subseteq N_Q$ of $F(Q_S)$, $F(Q''_S)$ and $F(Q'_S)$, and the identity for Q follows from the identity for the Q_S, which holds by induction for $S \neq N_Q$ and, for $S = N_Q$, whose saddlepoint lies in the closed orthant with j-th component zero, is $\hat P' - \hat P'' = \hat P$ above. In dimension one $F(Q') = 1$, and the three cases are the definition, the definition applied to Q'', and the reflection at the mean of Corollary 2. Continuity now follows, again by induction on d. Near a point $\bar x^*$ let Z be the set of components of $\hat s$ that vanish there and N^- the set of those that are negative. On each region near $\bar x^*$ on which the signs of $\hat s_Z$ are constant, $F(Q)$ is the alternating sum over the subsets of its negative set $N^- \cup Z'$, $Z' \subseteq Z$, whose full-dimensional term is the expansion of Theorem 1 at a saddlepoint in the closed orthant, continuous up to $\bar x^*$ by the continuity of the $I(\nu)$ there, and whose other terms are of lower dimension and continuous by induction; its limit at $\bar x^*$ is the alternating sum over $N^- \cup Z'$ of the $F(Q_S(\bar x^*))$, which is $F(Q(\bar x^*))$ by the consequence of the identity, whatever Z'. So F is continuous wherever it is defined, and condition (a) plays no part in this; what (a) supplies, besides the error bound, is the existence of the saddlepoints, which the recursion otherwise has to assume.

The proof of Proposition 1 uses the exponent of a normal orthant probability with thresholds of order $\sqrt{n}$.

LEMMA 5. *Let Σ be positive definite, $t \in \mathbb{R}^d$, $Z \sim N(0,\Sigma)$ and $q(t) = \min\{z^\top\Sigma^{-1}z : z \geq t\}$. Then $n^{-1}\log\mathbb{P}(Z \geq \sqrt{n}t) \to -q(t)/2$.*

PROOF. Let z^* attain the minimum, which is unique since $\{z \geq t\}$ is closed and convex. For $z \geq t$ the variational inequality $(z - z^*)^\top\Sigma^{-1}z^* \geq 0$ gives $z^\top\Sigma^{-1}z^* \geq q$, so with $\lambda = \sqrt{n}\,\Sigma^{-1}z^*$, for which $\lambda^\top Z \sim N(0, nq)$, $\mathbb{P}(Z \geq \sqrt{n}t) \leq \mathbb{P}(\lambda^\top Z \geq nq) \leq e^{-nq/2}$ when $q > 0$, and the bound is trivial when $q = 0$. For the lower bound let $B = z^* + [0,\delta]^d \subseteq \{z \geq t\}$; then $\mathbb{P}(Z \geq \sqrt{n}t) \geq \int_{\sqrt{n}B}\phi(w,\Sigma)\,\mathrm{d}w \geq (\sqrt{n}\delta)^d \min_{z\in B}\phi(\sqrt{n}z,\Sigma) \geq Cn^{d/2}\delta^d\exp\{-n\max_{z\in B}z^\top\Sigma^{-1}z/2\}$, and $\max_{z\in B}z^\top\Sigma^{-1}z \leq q + C'\delta$; let $n\to\infty$ and then $\delta \to 0$. □

PROOF OF PROPOSITION 1. Since $\hat s_1 < 0 \leq \hat s_2$, $N = \{1\}$, condition (a) of Theorem 2 is $\tilde s_2(0) \geq 0$ together with $\hat s_2 \geq 0$, and (b) follows from (a), so $\mathbb{P}(\bar X \geq \bar x) = P_\emptyset\{1 + O(e^{-cn})\}$ with $P_\emptyset = \mathbb{P}(\bar X_2 \geq \bar x_2)$, and Theorem 1 with $p = 0$ for the marginal of X_2, whose saddlepoint is $\tilde s_2(0) \geq 0$ and whose exponent $K(0,\tilde s_2(0)) - \tilde s_2(0)\bar x_2$ is $\ell(0)$, gives $n^{-1}\log\mathbb{P}(\bar X \geq \bar x) \to \ell(0)$. For the leading term, $E_nI(0) = e^{n(\hat K - \hat s^\top\bar x + \hat s^\top\hat K''\hat s/2)}\mathbb{P}(Z \geq \sqrt{n}\hat K''\hat s)$ with $Z \sim N(0,\hat K'')$, so by Lemma 5

$$\frac{1}{n}\log E_nI(0) \to \ell(\hat s_1) + \tfrac12\hat s^\top\hat K''\hat s - \tfrac12 q(\hat K''\hat s) = \ell(\hat s_1) + \max_{\mu\geq 0}\{-\mu^\top\hat s - \tfrac12\mu^\top(\hat K'')^{-1}\mu\},$$

by the substitution $z = \hat K''\hat s + \mu$ in q. The maximum is the Lagrange dual of the convex quadratic program $\min_{s\geq 0}\frac12(s-\hat s)^\top\hat K''(s-\hat s)$, whose value it equals by strong duality, and

that program is the minimization over the closed positive orthant of the quadratic approximation of $K(s) - s^\top \bar{x}$ about $\hat{s}$. Its minimizer over the face $s_1 = 0$ is $s_2 = \hat{s}_2 + \hat{K}''_{12}\hat{s}_1/\hat{K}''_{22}$, positive by hypothesis, where the derivative with respect to s_1 equals $-\hat{s}_1(\hat{K}''_{11} - \hat{K}''^2_{12}/\hat{K}''_{22}) > 0$, so the Karush–Kuhn–Tucker conditions hold there and the value of the program is $\frac{1}{2}\hat{s}_1^2\ell''(\hat{s}_1)$, with $\ell''(\hat{s}_1)$ the Schur complement by the implicit function theorem applied to the minimizer $\tilde{s}_2(s_1)$ near $\hat{s}_1$. Hence the limit in (23) is $\ell(\hat{s}_1) + \frac{1}{2}\hat{s}_1^2\ell''(\hat{s}_1) - \ell(0) = -\Delta/n$, and the mean-value form follows from Taylor's theorem for ℓ on $[\hat{s}_1, 0]$, where ℓ is three times differentiable by the implicit function theorem since the minimizer is attained, with $\ell'(\hat{s}_1) = 0$. Finally, by (14) and (15) and the scaling $\mathrm{He}_\nu(\sqrt{n}x, n\Sigma) = n^{-|\nu|/2}\mathrm{He}_\nu(x, \Sigma)$, $n^{|\nu|/2}I(\nu)$ at $\hat{s}$ is the integral over $\{w \geq 0\}$ of $\mathrm{He}_\nu(w, \hat{K}'')\phi(t_n + w, \hat{K}'')$, $t_n = \sqrt{n}\hat{K}''\hat{s}$, a polynomial of degree $|\nu|$ in w times the Gaussian whose mass on the orthant is $I(0)$. Split the orthant at $|w| = M\sqrt{n}$: on $|w| \leq M\sqrt{n}$ the polynomial is bounded by $Cn^{|\nu|/2}$, so this part is at most $Cn^{|\nu|/2}I(0)$; on $|w| > M\sqrt{n}$ the integral is at most $C\exp\{-\lambda(M - |\hat{K}''\hat{s}|)^2 n/4\}$ for $M > |\hat{K}''\hat{s}|$, while the cube argument of Lemma 5 with side $n^{-1/2}$ gives $I(0) \geq c\exp(-nq/2 - C'\sqrt{n})$, so that for M large this part is exponentially smaller than $I(0)$. Hence $|I(\nu)| \leq CI(0)$ and $E_n n^r |c_{r,\nu} I(\nu)| \leq Cn^r E_n I(0)$. □

PROOF OF THEOREM 3. The cumulant generating function of $\bar{X}$ is $nK(t/n)$, so by Fourier inversion along $\mathrm{Re}\, s = \hat{s}$ and the substitution $s = \hat{s} + \mathrm{i}y/\sqrt{n}$, by (3),

$$f_{\bar{X}}(\bar{x}) = \frac{n^{d/2} e^{n(\hat{K} - \hat{s}^\top \bar{x})}}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-y^\top \hat{K}'' y/2} \exp\{nG(\mathrm{i}y/\sqrt{n})\}\, \mathrm{d}y. \tag{53}$$

The integrand in (53) has modulus $|M_{\hat{s}}(y/\sqrt{n})|^n$, with $M_{\hat{s}}$ the tilted characteristic function of Section 2. By (1), $\rho = \sup\{|M_{\hat{s}}(u)| : |u| \geq \delta, \hat{s} \in T\} < 1$, and for $n \geq n_0$ the part of (53) with $|y| > \delta\sqrt{n}$ is at most $n^{d/2}\rho^{n-n_0} \sup_{\hat{s}\in T} \int_{\mathbb{R}^d} |M_{\hat{s}}(u)|^{n_0} \mathrm{d}u$ times the prefactor, exponentially small relative to the leading term. On $|y| \leq \delta\sqrt{n}$ replace $\exp\{nG(\mathrm{i}y/\sqrt{n})\}$ by the sum in (52) with $p = 2M - 1$: by Lemma 3, the remainder contributes at most $Cn^{-M} \int_{\mathbb{R}^d} |y|^{2M} e^{-\lambda|y|^2/4} \mathrm{d}y$ times the prefactor, and extending the integrals of the retained terms to the whole space changes them by exponentially small amounts. Integrating the retained terms against $(2\pi)^{-d}\exp(-y^\top \hat{K}'' y/2)$ gives $(2\pi)^{-d/2}|\hat{K}''|^{-1/2}$ times $\mathrm{i}^{|\nu|}E(S^\nu) = \mathrm{He}_\nu(0, \hat{K}'')$, where $S \sim N(0, (\hat{K}'')^{-1})$, by the Fourier representation of $\phi(x, \Sigma)$. The terms with $|\nu|$ odd, hence with m odd, vanish, so the surviving terms have $|\nu| = 2m + 2r$ with $m \leq M - 1$ and $r \leq 2m$. All bounds are uniform for $\hat{s} \in T$. □

APPENDIX C: COMPUTATION

The coefficients $c_{r,\nu}$ for all r and $|\nu| \leq N$ are stored in vectors indexed by ν, with the multi-indices ordered first by $|\nu|$, then by the partial sums $\nu_1 + \cdots + \nu_{d-1}$, and so on (Hillier, Kan and Wang, 2009, Lemma 2). With $F(i, j) = \binom{i+j}{j}$ the number of j-vectors with sum at most i, and $F(-1, j) = 0$, the position of ν is $\sum_{i=1}^d F(s_i(\nu) - 1, i) + 1$, $s_i(\nu) = \nu_1 + \cdots + \nu_i$, and the position of $\nu + e_j$ within the next block is that of ν plus $\sum_{i=j}^{d-1} F(s_i(\nu), i - 1)$, which is independent of $|\nu|$; these displacements are computed once. The recursion (5) is then a loop over r, over $|\nu|$ in decreasing order (so that one vector can be overwritten in place), and over the $\lambda \leq \nu$ with $3 \leq |\lambda| \leq |\nu| - 3(r - 1)$. The derivatives $\hat{K}_\nu$ at the saddlepoint are required for $|\nu| \leq 2M$ (density of order M) or $|\nu| \leq p + 2$ (distribution function with terms $m \leq p$). For the examples of Section 5 they are available in closed form, through the trace recursion (43) for the quadratic forms and from explicit cumulant generating functions otherwise; for a general cumulant generating function they can be obtained by automatic differentiation.

For the distribution function the $J(r)$ follow from Lemma 2 in increasing order of the number of components equal to -1, and the $I(\nu)$ from (14). Both steps are exact, but in floating-point arithmetic they lose accuracy by cancellation when $\sqrt{n}\hat{K}''\hat{s}$ has large components and $|\nu|$ is large: the terms of the binomial sum (14) are then much larger than $I(\nu) = O(n^{-|\nu|/2})$. In one dimension the same occurs in the recursion (35): the $I_m(u)$ are its minimal solution, and the forward recursion loses about $\log_{10}\{u^{2m+1}/m!\}$ digits at index m, all of them at the points of the singly noncentral F grid of the Supplementary Material with $n \le 2$, where $|\hat{u}_n|$ is large and the θ_k are large. Evaluating (34) on its contour, $v = u + \mathrm{i}y$, gives, for $u > 0$,

$$I_{2k}(u) = (-1)^k \phi(u)\, u\, M_k(u), \qquad I_{2k+1}(u) = (-1)^k \phi(u)\, M_{k+1}(u),$$

where $M_j(u) = E\{Z^{2j}/(u^2 + Z^2)\}$, $Z \sim N(0,1)$, are expectations of positive functions. Our programs use (35) for $|u| \le 2.5$ and otherwise evaluate the $M_j(u)$ by 80-point Gauss–Hermite quadrature, which converges exponentially because the integrand is analytic in the strip $|\mathrm{Im}\, z| < u$; the relative error is below 2×10^{-15} for $m \le 33$. In the same way, evaluating (11) on its contour, $s = \hat{s} + \mathrm{i}y$, gives

$$I(\nu) = \phi(t, \Sigma)\, \mathrm{Re}\, E\Big\{ \frac{(\mathrm{i}W)^\nu}{\prod_{j=1}^d (\hat{s}_j + \mathrm{i}W_j)} \Big\}, \qquad W \sim N(0, \Sigma^{-1}), \quad \Sigma = n\hat{K}'', \quad t = \Sigma\hat{s},$$

which requires neither Lemma 2 nor (14) nor any orthant probability. In two and three dimensions the double-precision programs use this representation, with 80×80 or 48^3 Gauss–Hermite nodes, when $\hat{s}_j\{(\Sigma^{-1})_{jj}\}^{-1/2} > 2.5$ for all j, and otherwise Gauss–Legendre quadrature of the orthant-integral form of $I(\nu)$ in conditioned coordinates, with the quadrature error estimated by halving the number of nodes; for $d \ge 4$ they evaluate the orthant-integral form by randomized quasi-Monte Carlo integration after the sequential conditioning of Genz (1992) (see also Genz and Bretz, 2009, Chapter 4), with a standard error over random shifts in place of a deterministic error estimate; the density expansion, which needs no integrals, is unaffected by the dimension. The node counts, the treatment of nearly singular conditional correlations and the checks against 40-digit computations are given in the Supplementary Material. The same algorithm serves every order, but its cost grows with the order and the dimension: at order five in three dimensions there are $\binom{30}{3} = 4060$ integrals $I(\nu)$, and a point of Table 4 takes about a minute on a laptop.

A second source of cancellation lies in the expansion (12) itself: its terms $n^r c_{r,\nu} I(\nu)$ can be much larger than their sum. This occurs when the components are strongly correlated under the tilted distribution and the reflection of Section 3.3 places t far from the vertex of the orthant in one coordinate, as in the sample autocorrelations of the Supplementary Material with $\rho = 0.5$, where the correlation implied by $\hat{K}''$ is about 0.9 in absolute value. Our programs return $\mathcal{C}$, the ratio of the sum of the absolute values of the terms of (12), $I(0)$ included, to the absolute value of their sum; about $\log_{10}\mathcal{C}$ digits are lost to this cancellation in double precision, before the errors of the individual terms. Evaluated as written, at order five $\mathcal{C}$ is at most 2×10^6 at the points of Table 6 and 7×10^6 in the blocks with $\rho = 0$ of the autocorrelation table of the Supplementary Material, but it reaches 6×10^5 to 3×10^{15} in the block $N = 100$, $\rho = 0.5$ and 2×10^6 to 2×10^{17} in the block $N = 50$, $\rho = 0.5$, beyond the resolution of double precision.

The cancellation is a matter of coordinates, and it is largely removed by evaluating the terms for a whitened vector. In the computational formulas that follow, n is absorbed into K, as in Section 5, so that $\Sigma = \hat{K}''$ and $t = \Sigma\hat{s}$. Let $\hat{K}'' = LL^\top$ be the Cholesky factorization and $Y = L^{-1}X$. The orthant event $\{\bar{X} \ge \bar{x}\}$ is preserved by no linear map other than a positive diagonal scaling combined with a permutation of the coordinates, so the expansion cannot be applied to Y directly; but the sum of the terms of (12) with a given r and a given degree $|\nu|$ is a contraction of the tensor of coefficients $c_{r,\nu}$ with the tensor of integrals $I(\nu)$,

and is invariant under the change of coordinates applied to both. Since $\mathrm{He}_\nu(w,\Sigma)\phi(w,\Sigma) = (-\partial_w)^\nu\phi(w,\Sigma)$ and $\partial_w = L^{-\top}\partial_v$ for $w = Lv$, where $L^{-\top} = (L^{-1})^\top = (L^\top)^{-1}$, the polynomial $\sum_{|\nu|=k} c_{r,\nu}\mathrm{He}_\nu(w,\hat{K}'')$ equals $\sum_{|\mu|=k}\tilde{c}_{r,\mu}\prod_{j=1}^d \mathrm{He}_{\mu_j}(v_j)$, where the $\tilde{c}_{r,\mu}$ are the coefficients (4) built from the Taylor coefficients of $K_Y(u) = K(L^{-\top}u)$ at $L^\top\hat{s}$, the cumulant generating function of Y, whose Hessian there is the identity. Hence

$$\sum_{|\nu|=k} c_{r,\nu} I(\nu) = \sum_{|\mu|=k}\tilde{c}_{r,\mu}\tilde{I}(\mu),$$

$$\tilde{I}(\mu) = \int_{t_1}^{\infty}\cdots\int_{t_d}^{\infty}\prod_{j=1}^{d}\mathrm{He}_{\mu_j}\{(L^{-1}(z-t))_j\}\,\phi(z,\hat{K}'')\,\mathrm{d}z_1\cdots\mathrm{d}z_d, \tag{54}$$

an integral over the same orthant, with the same quadrature nodes as before and the univariate Hermite polynomials of the whitened variable in place of $\mathrm{He}_\nu(z-t,\hat{K}'')$; in the Gauss–Hermite representation above the same substitution replaces $(\mathrm{i}W)^\nu$ by $(\mathrm{i}V)^\mu$ with $W = L^{-\top}V$ and V standard normal. The terms of a given order are not individually invariant, only their sum is, and in the whitened form they cancel much less: at order five $\mathcal{C}$ falls to at most 4 at the points of Table 6, to at most 6×10^2 in the blocks with $\rho = 0$ of the autocorrelation table of the Supplementary Material and to 2 to 4.5×10^4 in the blocks with $\rho = 0.5$. In the bivariate examples of Sections 5.1 and 5.2 the gain is of the same kind: at order five, $\mathcal{C}$ in the original coordinates is 1.1×10^7, 2.3×10^9 and 5.1×10^9 at the points $(3.0,3.0)$, $(3.5,4.0)$ and $(4.0,4.0)$ of Table 2 and 4.6×10^8 at the point $(2.0,2.0)$ of Table 1, against 42, 79, 1.1×10^2 and 8 in the whitened form, and at most 1.2×10^2 at every point of Tables 1, 2 and 3. The density expansion (24) is whitened in the same way, with $\mathrm{He}_\mu(0,I) = \prod_{j=1}^d \mathrm{He}_{\mu_j}(0)$ in place of $\mathrm{He}_\nu(0,\hat{K}'')$, and the whitening applies to the quadrature and quasi-Monte Carlo evaluations of $\tilde{I}(\mu)$ in any dimension, but not to the recursion of Lemma 2, which is tied to the orthant. Two details of the implementation, the direct computation of the Taylor coefficients of K_Y and a separate estimate of the rounding error of the coefficients, are described in the Supplementary Material; with these, the double-precision expansions agree with independent computations in 40- and 50-digit arithmetic to 7×10^{-14} at every point of Table 6 and to 1.3×10^{-12} at every point of the autocorrelation table of the Supplementary Material. Our programs evaluate all multivariate expansions in the whitened form.

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SUPPLEMENTARY MATERIAL

Supplement to "Uniform saddlepoint expansions for multivariate tail probabilities" The Lugannani–Rice expansion in the notation of the paper, with proofs; the comparisons of the univariate approximations of orders one to six with exact values for the noncentral chi-squared and the singly and doubly noncentral F distributions; details on the approximations of Wang (1990) and of Kolassa and Li (2010); the sample autocorrelations at two lags; computational details; and a zip archive containing the MATLAB programs that implement the expansions, the scripts and data that reproduce the tables, and a user manual.

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